

The Double Measurement Confound in Agent Benchmarks: De-Scaffolding, Ground-Truth Scoring, and Reliability Beyond the Mean

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Abstract

Agent benchmarks are increasingly used to compare large language models (LLMs) and guide deployment decisions, yet benchmark scores are meaningful only if they measure model capability rather than properties of the evaluation pipeline. We identify a *double measurement confound*: execution-critical decisions are performed by a fixed scaffold instead of the model, while the scorer evaluates outputs using criteria that may not reflect task correctness. We unify these issues within a measurement-theoretic framework that characterizes when benchmark scores can be interpreted as evidence of model capability, and instantiate it with an audit-and-repair protocol that (i) transfers execution-critical decisions from the scaffold to the model, (ii) replaces shape-based evaluation with seeded ground-truth scoring, and (iii) reports reliability beyond the mean through worst-case and tail-risk metrics. Experiments on COMTRADEBENCH show that the joint intervention transforms a nearly flat leaderboard into a reliability spectrum that distinguishes both average performance and robustness across seeds. Applying the audit to existing benchmarks further shows that scorer validity is benchmark-specific, whereas scaffold ownership is an uncontrolled axis wherever we probed it. Our results suggest that benchmark scores should be interpreted together with their scaffolding level, scoring criterion, and reliability profile, providing a practical framework for more valid evaluation of LLM agents.

1 Introduction

Benchmarks for large language model (LLM) agents evaluate a complete execution system that includes the model, tools, scaffold, environment, and scorer. The reported score may therefore depend on the evaluation procedure as well as the underlying model. Prior work has separately shown that harness choices can change agent performance (Yao et al. 2026; Kapoor et al. 2026), automatic evaluators can disagree with task correctness (Vaghasiya et al. 2026; Xiao et al. 2023), and aggregate success rates can conceal unreliable behavior (Yao et al. 2025; Rabanser et al. 2026). What remains unclear is how these measurement problems interact and when an agent benchmark can support claims about model capability.

This issue is pronounced on a published benchmark for adversarial, trade-data-shaped tool use. Two frontier models from independent labs, Kimi and Claude, obtain identical

mean scores of 97.5, while a rule-based script with no LLM scores 96.8. Two unrelated open models, Qwen2.5-7B and Llama-3.3-70B, also receive byte-identical per-seed scores. At this configuration, the benchmark has little power to distinguish model capability (Raji et al. 2021; Bowman and Dahl 2021; Liao et al. 2021; Reuel et al. 2024).

We trace this result to a *double measurement confound* involving the scaffold and the scorer. The **scaffold** makes every execution-critical decision, including retry, pagination, deduplication, and submission. The score therefore reflects execution performed by the harness rather than the model. The **scorer** evaluates output shape and self-reported metadata instead of comparing the submitted values with ground truth (Messick 1995; Jacobs and Wallach 2021; Xiao et al. 2023). Consequently, a fabricated record set receives the same score as the correct one, 0.987 in both cases. These artifacts also conceal one another. A valid scorer still measures decisions made by the scaffold, while a model-controlled execution remains mismeasured by an invalid scorer. We formalize this interaction as a joint identification problem.

We transfer the execution-critical decisions to the model and score its submission by ground-truth F_1 (Section 4.3). This joint intervention turns eight previously indistinguishable models into a graded spectrum, from Claude-Fable-5 at 0.974 to Llama-3.3-70B and Qwen2.5-7B at 0.000 (Figure 2). A paired seven-model 2×2 intervention shows that neither change is sufficient alone. Under the production scaffold, payloads are byte-identical across all 21 model pairs, and both scorers tie every model. The spectrum appears only when the model controls execution and the submission is scored against ground truth (Table 1). The benchmark pre-dates this audit: its scaffold, scorer, and leaderboard were fixed before the double-confound hypothesis, making this a retrospective self-audit rather than a benchmark built to exhibit the pathology.

The repaired evaluation also reveals reliability differences hidden by the mean (Henderson et al. 2018; Rabanser et al. 2026). GPT-4o ranks fifth by mean but reaches a worst-case score of 0.00, whereas the smaller Claude-Haiku-4.5 never falls below 0.787. Our pre-registered stress test (Nosek et al. 2018) is confirmed under its designated scorer, but the effect

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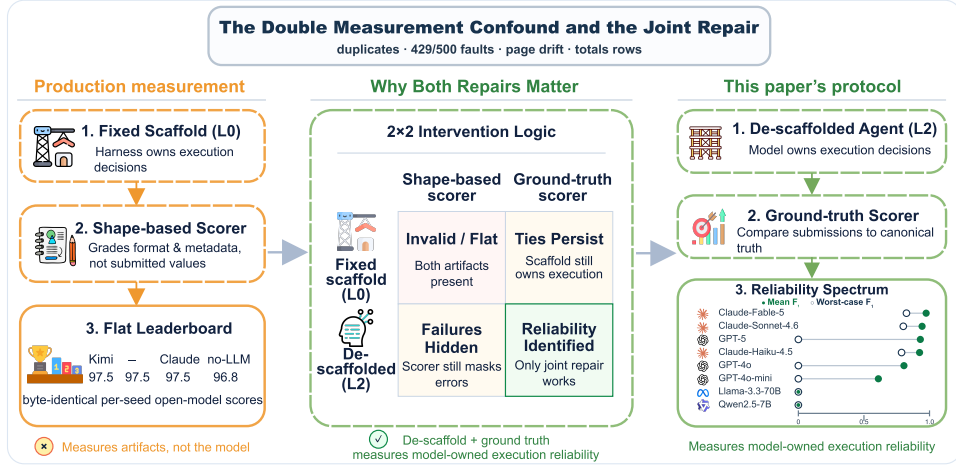


Figure 1: The double measurement confound and the joint repair. *Left* (production): a fixed scaffold (L0) owns the execution-critical decisions and a shape-based scorer grades format and metadata, so a no-LLM script ties the frontier. *Center*: why the two artifacts must be removed jointly (Section 3.3). *Right* (our protocol): a de-scaffolded agent (L2, Section 4.2) scored as ground-truth F_1 (Section 4.3) against seeded ground truth, resolving Figure 2. Schematic; every quoted number appears in the text.

disappears when the same episodes are evaluated against ground truth.

In external audits, the official τ -bench scorer is outcome-based, yet we find its four native scaffolds shift the same model’s reward by up to 0.267 and change model rankings (Yao et al. 2025). The Berkeley Function-Calling Leaderboard (BFCL) instead offers an abstract syntax tree (AST) track that scores a call against a function specification rather than execution success (Patil et al. 2025).

We make three contributions. First, we identify the *joint* measurement confound created by scaffold ownership and scorer validity, show that the two artifacts mask each other so that repairing the scorer alone leaves every model tied while de-scaffolding alone leaves the differences misgraded, and state the condition under which an agent score identifies model behavior. Second, we introduce BENCHAUDIT, an executable audit-and-repair protocol whose deliverable is a per-benchmark *validity card*, and demonstrate it on three benchmarks; applied reflexively, it overturns the result of our own pre-registered stress test. Third, we release COMTRADEBENCH as a seeded generator card together with the eight-model reliability spectrum, the paired 2x2 intervention, and the τ -bench and BFCL audits that reproduce every number.

2 Related Work

A benchmark score is evidence about a model only when three things hold. The model, not the harness, makes the execution-critical decisions. The scorer responds to whether the work is correct, not only to how it looks. The report shows how scores vary across runs, not only their mean. Prior work has largely developed each of these separately. We review four bodies of work below, one per condition plus documented scaffold sensitivity, and close with the non-stationary reinforcement-learning (RL) literature our apparatus inverts.

Across the seven suites in our positioning matrix (supplementary material) we find none that both reports seeded worst-case reliability and treats the scaffolding level as a controlled variable: τ -bench’s pass^k covers the first without seeded worst-case estimation, and Harness-Bench’s 6x8 factorial (Yao et al. 2026) covers the second over complete harnesses rather than a decomposed ownership axis.

Tool-use and agent benchmarks. ToolBench (Qin et al. 2024), BFCL (Yan et al. 2024; Patil et al. 2025), API-Bank (Li et al. 2023), AgentBench (Liu et al. 2024), τ -bench (Yao et al. 2025), SWE-bench (Jimenez et al. 2024), and GAIA, WebArena, and OSWorld (Mialon et al. 2024; Zhou et al. 2024; Xie et al. 2024) share two properties we build against. First, their adversity, where present, lives in the prompt or task specification; COMTRADEBENCH places it in the environment’s response dynamics, where it cannot be prompted away (τ -bench is the closest relative, but its adversity is rule- and policy-level, not environment-level fault injection with within-episode dynamics). Second, they report means (or best-of- k pass@ k) over few runs, and none treats the *scaffolding level* as controlled. SWE-bench scores move substantially when the harness changes while the model is held fixed (Yang et al. 2024; Xia et al. 2025); holistic leaderboards run the same models across several scaffolds (Kapoor et al. 2026); and Harness-Bench (Yao et al. 2026) runs a 6x8 harness-model factorial while explicitly *not* decomposing individual harness mechanisms. We formalize that missing decomposition into a controlled *ownership* axis, with seeded repeats that support worst-case reporting.

Reliability and worst-case evaluation. We import a tail-risk vocabulary under-applied to LLM-agent evaluation: CVaR from finance (Rockafellar and Uryasev 2000), carried into risk-sensitive and distributional RL (Chow et al. 2015; Bellemare, Dabney, and Munos 2017); worst-group accuracy

from distributionally robust learning (Sagawa et al. 2020); and multi-seed variance and confidence-interval reporting (Agarwal et al. 2021). Two points sharpen our position: the worst case *reorders* the mean ranking up to the frontier (a cross-model *re-discriminator*, not only a within-model tool), and where worst-group robustness conditions on known sub-populations, we condition on seeds of a seeded adversary, making the worst case estimable rather than anecdotal.

Benchmark validity and automatic scorers. A separate line asks whether a benchmark score means what it is read to mean: construct and criterion validity in measurement theory (Messick 1995; Campbell and Fiske 1959), its transfer to machine learning (Jacobs and Wallach 2021; Hutchinson et al. 2022), benchmark-level critiques (Raji et al. 2021; Bowman and Dahl 2021; Liao et al. 2021; Reuel et al. 2024), and analyses of automatic evaluation metrics and prompt-dependent scores (Xiao et al. 2023; Mizrahi et al. 2024). This work establishes that scores can fail to track the construct; it does not ask *who executed* the task being scored. We treat scorer validity and scaffold ownership as a single identification problem, and make the resulting condition auditable rather than arguable.

LLM-agent variance and scaffolding sensitivity. LLM agents are documented as high-variance across seeds, prompts, and harnesses (Miller 2024; Kapoor et al. 2025; Sclar et al. 2024), and the scaffold can shift measured success as much as the model (Yang et al. 2024; Xia et al. 2025). Prior work *observes* the variance as a confound; we make it the *object of measurement* via a controlled scaffolding spectrum, trace a large share of the apparent variance to the scaffold’s hidden execution work (Section 3), and show the diagnosis is *correctable*.

Non-stationary and continual RL. Concept drift (Gama et al. 2014), non-stationary bandits (Besbes, Gur, and Zeevi 2014), continual RL (Khetarpal et al. 2022), and hidden-mode POMDPs (Choi, Yeung, and Zhang 2000) study environments that change while a learner adapts. Our apparatus inverts the object: the agent is *frozen* and the question is whether a fixed policy executes reliably under within-episode adversity, with escalation driven by agent success rather than elapsed time (Lee et al. 2020).

3 The Double Measurement Confound

COMTRADEBENCH is a seeded benchmark of adversarial, paginated data extraction, themed on public trade-statistics records: given a query and an operational budget, an agent must fetch all pages of a mock data API, deduplicate records, drop summary rows, retry transient faults, and submit a cleaned record set with metadata and a run log. The adversity is *in the environment* (faults, duplicates, and distractor rows are injected by the mock service, not described in the prompt), so it cannot be reasoned around. Scope: COMTRADEBENCH is stylized adversarial extract-transform-load work themed on trade data; it tests execution reliability, not economic reasoning. All data is a deterministic function of `(task_id, seed)`, so ground truth is regenerable from the seed (Section 4).

Ten tasks isolate fault families, with a fault-free variant (`T9_clean_ab`) as the substrate for the A/B design of Section 5; per-task parameters are in the supplementary material. The agent interacts through three model context protocol (MCP) tools, constant across all scaffolding levels: `get_task_info()`, `fetch_page(page, page_size)`, and `submit_results(data, metadata, run_log)`. The tools are fixed; what varies is the *wrapper* around them: the first artifact.

3.1 Artifact 1: The Scaffold Does the Work

In the production scaffold, the agent loop wraps the tool calls in fixed control logic: it auto-retries transient faults with backoff, deduplicates fetched rows by primary key, drops totals rows, loops pagination to completion, and assembles and submits the final payload. These are the execution-critical decisions the adversarial API is designed to stress. When the scaffold owns them, the LLM only fills templated slots (which query to run, when to stop paging).

The resulting invariance is not approximate. On the published leaderboard, Kimi and Claude post numerically identical mean scores (97.5 each), and a rule-based baseline with no LLM at all scores 96.8, within roughly 0.7 points of the frontier agents. At the seed level the invariance is exact: under the production agent, Qwen2.5-7B and Llama-3.3-70B produce *byte-identical per-seed reward vectors* on the multi-page task.

3.2 Artifact 2: The Judge Scores Shape

The benchmark’s deployed scorer (`judge.py`) is a deterministic, 100-point rubric over six dimensions: correctness (30), completeness (15), robustness (15), efficiency (15), data quality (15), and observability (10), with two governance gates. This is a criterion-validity question: *the judge scores output shape and self-reported metadata, never the submitted values against ground truth*. Correctness grades the submitted-line *count* against the submitter’s own `metadata.row_count` (never whether the ids are the true ids), robustness rewards *logging* a retry rather than retrieving records, and no dimension compares the submission to the seeded ground truth (full per-dimension walk-through, gates included, in the supplementary material). A probe in the supplementary material passes three canned submissions through the released `judge.py` and reports each six-dimension breakdown: an empty `data.jsonl` with well-formed metadata and a disciplined run log scores 0.648, and a fabricated record set with invented ids and self-consistent metadata receives the *identical* judge total as the correct ground-truth record set (0.987 each, at ground-truth F_1 0.0 versus 1.0). Metadata-shape and log-discipline credit floor a contentless submission at one half or above, while the rubric compresses real differences at the top. The released pipeline’s probe battery (Section 4) adds the converse test: the *same* correct records paired with a contradictory self-report drop by 0.398, while fabricating the data behind a well-formed report costs 0.000. The judge grades the report, not the data.

3.3 Why the Artifacts Must Be Removed Jointly

The two artifacts mask each other (Figure 1). Fix the scorer alone and it faithfully reports the scaffold’s near-perfect execution: the model still owns no decision the adversary targets, so the model-invariant ties persist. Remove the scaffold alone and the shape-based judge hides the model’s real failures on the way out, flooring the empty submission, ceilinging the perfect one, and compressing the genuine spread into a band that a leaderboard reads as noise. Only joint removal recovers a measurement of the model: return retry, deduplication, totals filtering, and the submit decision to the model, and score the submitted unique records against the seeded ground-truth record set, with duplicates and totals rows counted as errors; the seeded generator makes this scorer free and deterministic per `(task, seed)`. Section 4 specifies the resulting protocol. We keep the six-dimensional judge in the released benchmark as its default scorer and as an object of study: the judge–ground-truth contrast is itself the evidence for the measurement-validity claim. Section 5 tests this as a 2×2 intervention (Table 1).

4 Measurement Protocol

We now specify the protocol; the supplementary material diagrams the full pipeline alongside the measured L0/L1/L2 spectrum, and Figure 1 contrasts its scoring path with the production path.

4.1 A Measurement Model

Write a benchmark score as $S(m, c, j)$: the reward model m receives when run under scaffold c (a harness owning some subset of the execution-critical decisions of Section 3) and graded by scorer j . A leaderboard reports $S(\cdot, c_0, j_0)$ at one fixed configuration and interprets it as a property of m alone (Mizrahi et al. 2024). Call the configuration *model-invariant* if $S(m_1, c_0, j_0) = S(m_2, c_0, j_0)$ for all models under test, and call j *criterion-valid* if S is strictly increasing in the ground-truth quality of the submitted work product.

Proposition 1 (non-identifiability). If (i) the scaffold owns every execution-critical decision, so that no scored quantity depends on a model choice beyond templated slots, and (ii) the scorer grades only submission shape and self-reported metadata, then $S(m, c_0, j_0)$ is invariant in m whenever two models fill the template equivalently, and no function of the leaderboard identifies a property of the model. Both hypotheses and the conclusion hold at the production configuration (Section 3): the invariance is realized byte-for-byte across two unrelated open models. The protocol below is the identification strategy: move c to L2 so the decisions depend on m , and j to ground truth so the score depends on the decisions; dependence on m then reappears (Figure 2).

Proposition 2 (identification). Conversely, if every execution-critical decision is model-owned and the scorer is criterion-valid, then the score is a deterministic function of the model’s decisions in the seeded environment, so any difference in S between two models at matched seeds reflects a difference in model behavior, and seed-level statistics (worst case, CVaR) estimate the model’s execution reliability. What behavior is measured remains level-dependent: at

L2 the identified quantity includes the re-emission demand, the mechanical-floor caveat of Section 4.2.

The propositions yield a minimal condition a benchmark can be audited against. Write D_{claimed} for the decisions the benchmark’s construct claims to measure and D_{model} for the decisions its harness actually leaves to the model; the capability claim is interpretable only if $D_{\text{claimed}} \subseteq D_{\text{model}}$, and when the inclusion fails there are exactly two repairs: return the decisions to the model, or relabel the score as system-level performance (Hutchinson et al. 2022). Ownership is necessary, not sufficient: a model-owned channel must also be *non-degenerate*, with $\text{Var}(A) > 0$ over the chosen actions and $E[Y \mid A=a_1] \neq E[Y \mid A=a_2]$ for some action pair. Our L1 instrument fails exactly this check (SKIP chosen 0 of 120 episodes; Section 5.3), and no sample size repairs it: the effect is non-identified, not merely low-powered (Card et al. 2020). Rule-policy calibration (always-retry, always-skip, quota-aware) is the pre-run check.

4.2 The Scaffolding Spectrum

A benchmark evaluates a model *inside a harness*, and the harness makes the execution-critical decisions of Section 3. We call this bundle the *scaffold* and treat the share of it owned by harness versus model as a controlled variable with three reference levels.

L0: production auto-everything. The harness performs every execution-critical step; the model fills templated slots and owns no decision the adversary targets. L0 is a common shipping configuration; it maximizes mean score and minimizes model signal, producing the ties of Section 3.

L1: half-scaffold. The harness handles the mechanical bottleneck (pagination, writing the final record file); the model owns the one adversary-relevant decision: on a fault, retry or skip.

L2: full slim. The model owns everything, including re-emitting the cleaned record set and calling `submit_results`. L2 is where the populated reliability spectrum of Figure 2 appears.

L0 hides all model differences; L2 exposes the spectrum but, for the weakest models, folds “decided wrongly under adversity” and “failed the mechanical re-emission” into one score; L1 separates them, and shows the isolated decision to be degenerate here (Section 5): L0 and L2 are the informative extremes, L1 the diagnostic explaining the floored zeros. A number quoted without its scaffolding level is not interpretable.

4.3 The Ground-Truth Metric

The shipped judge does not support a reliability claim (Section 3). For all headline numbers we score against the seeded canonical record set. The generator emits each record with a deterministic id (`{task}-{i:06d}`), so the true unique-record set for any `(task, seed)` is known exactly. With S the submitted records, G the ground-truth set, and U the distinct ground-truth records in S , $R = |U|/|G|$, $P = |U|/|S|$, and $F_1 = 2RP/(R + P)$: bounded in $[0, 1]$ and deterministic per `(task, seed)`. Unlike the judge, it drops when the data is wrong: omitted real rows cut recall; duplicates, totals rows, and fabricated rows inflate $|S|$ but not $|U|$, cutting precision.

COMTRADEBENCH is issued as a *generator card*, a datasheet for a seeded procedural generator rather than a static corpus (Geburu et al. 2021). Because a run reproduces from only the task id and the seed, exact test-item memorization is less likely, though generator- and template-level contamination remain possible (Zhu et al. 2024; Oren et al. 2024; Golchin and Surdeanu 2024; Jacovi et al. 2023).

4.4 Reliability Metrics

The quantity that pages a human is not the mean but the worst case across the adversary’s seeds. Each run is one (level, seed) $\rightarrow$ terminal reward, where a level is an opaque difficulty knob. We report: *worst-case* (minimum over tested seeds, an estimate of the deployment floor); *CVaR@ α* (mean of the worst α -fraction, $\alpha = 0.2$ default) (Rockafellar and Uryasev 2000); *Reliability@ τ* (fraction of runs clearing threshold τ ; $\tau = 0.9$ default, also reported at 0.5), akin in spirit to *pass^k* (Yao et al. 2025); a *seeded bootstrap 95% CI* for the mean (2000 iterations, bit-reproducible) (Efron 1979); and, for paired A/B comparisons, the within-pair *Cliff’s δ* (dominance $[\#(\Delta_i > 0) - \#(\Delta_i < 0)]/n$, the effect-size companion of the sign-flip test) with the same seeded bootstrap CI (Cliff 1993). Across difficulty levels we summarize the degradation curve and the *reliability gap*: the static mean minus the hardest-level worst-case, the operative column of Section 5.

4.5 The Non-Stationarity Engine

A domain-agnostic engine generates the stress reproducibly and lets us test whether its *shape* matters, with per-step randomness seeded by `sha256(seed:step)`: non-stationary does not mean non-reproducible. It supplies the matched-mean escalating-versus-constant A/B design we pre-register (Section 5); its architecture and scales are in the supplementary material.

The protocol as a reproducible audit pipeline. The full protocol is implemented as BENCHAUDIT: given a benchmark’s harness, scorer, and seeded tasks, it emits the scaffold-ownership map, the forged-submission scorer probe (correct, empty, fabricated, duplicate-laden, contradictory-metadata), the 2×2 joint intervention (Table 1), de-scaffolded reliability statistics, and a per-benchmark *validity card* whose cells are pass, fail, or *not probed*. Agents play no role in scoring; every reported number originates in the deterministic, seeded core, and a self-audit against COMTRADEBENCH reproduces each finding as an executable regression check. Three optional, human-confirmed LLM-assisted layers *propose* rather than score (scaffold discovery, harness synthesis, probe design), gated by a number guard that rejects any narrated figure absent from the audit JSON: an auditor that is itself auditable.

5 Experiments

We report three results on COMTRADEBENCH, then audit two external benchmarks and summarize every target as a validity card. All headline numbers use the ground-truth scorer of Section 4.3. E1 runs all eight models at the full de-scaffold level L2 on the five execution tasks of Figure 2, 10 seeds per (model, task); the escalation experiments use the L1 half-scaffold with a ground-truth coverage reward, $n = 20$ seeds

per arm, paired by seed. Every interval is a seeded bootstrap and reproduces bit-for-bit.

5.1 The De-Scaffolded Reliability Spectrum

Removing both artifacts of Section 3 (the LLM owns retry, dedup, totals filtering, re-emission, and submission, scored as ground-truth F_1) opens the flat leaderboard into a populated spectrum (Figure 2).

The spectrum spans its full range (Figure 2). Claude-Fable-5 leads on *both* mean (0.974) and worst case (0.824), clearing every single-fault task and leaving discriminative headroom at the top; a *previous-frontier tier* (Claude-Sonnet-4.6, GPT-5, and the small Claude-Haiku-4.5) clusters at mean 0.92–0.94 but is no longer uniform in reliability. A *high-variance mid tier* (GPT-4o, GPT-4o-mini) pairs strong means with a worst case of 0.00, and a *floored tier* (Llama-3.3-70B, Qwen2.5-7B) scores 0.000 throughout. The floored zeros are a marshaling failure, not a decision-quality result: Llama fetches rows (≈ 0.83 coverage at L1) yet submits an empty payload at L2, and Qwen never submits a valid one, which is why the scaffolding level travels with every number.

The tail statistics of Section 4.4 separate models the mean does not. GPT-5 and Claude-Haiku-4.5 differ by 0.006 in mean ground-truth F_1 (0.930 versus 0.924) but by 0.079 in CVaR@0.2 (0.720 versus 0.799); at $\tau = 0.9$, Reliability@ τ is 0.720 for GPT-5, 0.600 for Claude-Haiku-4.5, and 0.480 for GPT-4o. A mean-only report would call the first two equivalent.

The headline is that *worst-case reliability reorders models the mean rates as capable*. GPT-4o ranks fifth by mean yet floors at worst case, through genuine bimodality: on dedup (T3) its per-seed F_1 is $[0, 0, 1, 0, 0, 0, 0, 1, 1, 1]$, and a forensic replication ($n=30$) closes the mechanism, the outcome fully determined by whether the model ever varies an off-by-one `fetch_page(page=0)` call (14/14 of solves, 0/16 of failures): identifiable error-recovery behavior, not seed noise. The reorder reaches the frontier: GPT-5 is bimodal on mixed faults, handing the suite’s largest gap (0.930) to a model the mean ranks third, while the small Claude-Haiku never drops below 0.787. Reliability tracks neither raw size nor mean score (full transcript taxonomy in the supplementary material).

5.2 Scaffolding Level as a Hidden Axis

The same environment, measured at the three levels of Section 4.2, returns three different verdicts. At L0 (production auto-everything) all model differences vanish into the ties of Section 3: the score measures the scaffold. At L2 the spectrum of Figure 2 appears, but with a mechanical floor: part of the floored-tier zero is the marshaling chore, not adversity. L1, which hands the model only the single adversary-relevant decision (on a fault, retry or skip), shows that even this decision is degenerate here: across 120 episodes, SKIP is chosen *zero* times, and coverage settles near 0.83, model-invariant, capped by duplicate-overwrite in the mock rather than by any model choice; the always-retry substrate of the pre-registered null below. The scaffolding level thus changes not just the magnitude but the *qualitative content* of what is measured.

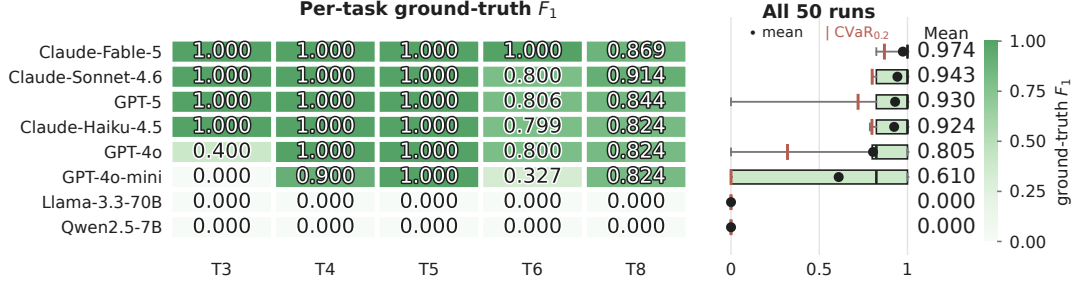


Figure 2: E1: the de-scaffolded (L2) reliability spectrum. *Left block*: per-task ground-truth F_1 (mean over 10 seeds) on the five execution tasks (T3 dedup, T4 429, T5 500, T6 drift, T8 mixed). *Right block*: all $n=50$ runs per model — box = interquartile range, whiskers = min-max, • = mean (also the numeric column), tick = CVaR at $\alpha=0.2$; rows sorted by mean. The lower whisker end is the worst-case, and it reorders the ranking the mean implies: GPT-5 and GPT-4o post high means (0.930, 0.805) yet whiskers reaching 0.00, while the far smaller Claude-Haiku-4.5 never drops below 0.787. Per-task dispersion, the reliability gap, and the excluded T7 diagnostic are in the supplementary material.

The audit pipeline runs the axis as a 2×2 *joint intervention*: one batch (T3_duplicates, five seeds per cell, seven of the eight models across three providers; Qwen’s provider was unavailable), every episode scored by both the shipped judge and ground truth, the harness swapped between L0 and L2 with nothing else changed. Table 1 reports the cell means (drawn as a 2×2 panel in the supplementary material). At L0 the submitted payloads are SHA-256-identical across *all 21 model pairs* on every seed, reasoning models included, so both scorers tie all seven: fixing the scorer alone reveals nothing. De-scaffolding alone still misgrades what it separates: among the models that submit, the judge compresses a 1.000 ground-truth spread to 0.425, scoring Llama’s all-wrong submissions above zero. Only the joint cell recovers the spectrum (0.000 to 1.000), so the masking argument of Section 3.3 is measured, not argued. Llama is served by a different provider here than in Figure 2 (BF16 versus FP8, disclosed) and floors regardless: the L2 floor is not a serving-stack artifact.

5.3 The Confound Inside the Confirmatory Experiment: a Pre-Registered Stress Test

This experiment closes the causal chain: the scorer half of the confound resurfaces inside our own pre-registered test. We pre-registered the hypothesis that escalating, progress-driven stress degrades agents more than matched-mean constant stress (PREREGISTRATION.md, frozen 2026-06-17) (Nosek et al. 2018), and report it as scorer-confound evidence, not a non-stationarity finding: it is confirmed under the scorer the pre-registration named and vanishes under ground truth. The B2 follow-ups are *exploratory*, *not pre-registered*: the enforced-quota variant the pre-registration explicitly deferred.

B1: a null, with a disclosed instrument deviation. On T9_clean_ab the matched-mean design is null for all three models ($\Delta = +0.000$; Cliff’s $\delta = +0.05$, 95% CI $[-0.30, +0.40]$), and SKIP, the only channel a model could move its score through, fired in 0 of 120 episodes;

Model	L0 (production)		L2 (slim)	
	judge	gt F_1	judge	gt F_1
Claude-Fable-5	0.987	1.000	0.902	1.000
Claude-Sonnet-4.6	0.987	1.000	0.948	1.000
GPT-5	0.987	1.000	0.768	1.000
Claude-Haiku-4.5	0.987	1.000	0.781	1.000
GPT-4o	0.987	1.000	0.614	0.400
GPT-4o-mini	0.987	1.000	0.000	0.000
Llama-3.3-70B	0.987	1.000	0.523	0.000
Model spread	0.000	0.000	0.948	1.000

Table 1: The 2×2 joint intervention: harness swap $\times$ scorer swap on shared episodes (T3_duplicates, $n=5$ seeds per cell, one batch, seven models over three providers; each episode double-scored; model order as Figure 2). L0 submissions are sha-256-identical across all 21 model pairs on all five seeds. GPT-4o-mini’s L2 zeros are non-submission (both scorers agree); among submitting models the judge compresses the 1.000 ground-truth spread to 0.425. Models separate fully only when both artifacts are removed (right-most column).

the run used the L1 coverage instrument, blind to the pre-registered dedup/totals channel (disclosed per the deviation rule; powered against large effects only, see Limitations). The pre-registration-faithful rerun (slim agent, terminal judge, $n=20/\text{arm}$) is null for Llama and Claude but gives GPT-5 $\Delta = +0.011$, CI $[+0.002, +0.023]$, $p = .034$, $\delta = +0.25 [-0.15, +0.65]$: a *confirmation* under the pre-registration’s “ ≥ 1 frontier reasoning model” outcome. We do not dismiss it as noise. The scorer explains it: on the identical episodes GPT-5’s ground-truth F_1 and coverage deltas are ≈ 0 and the submitted data is equally correct across arms, so the confirmation lives *entirely* in judge dimensions ground truth does not measure: the scorer-dependence this paper diagnoses, surfacing inside our own confirmatory experiment (results_b1_slim.json).

B2: an exploratory follow-up. B2 enforces a request quota with a one-shot mid-episode squeeze only the escalating arm crosses; the penalty turns positive on all three models (Llama and GPT-5 survive Holm correction, Claude does not) and revives the SKIP channel (0/120 $\rightarrow$ 38/120). But a no-LLM *always-retry* policy reproduces the flip at $\Delta = +0.096$, larger than every model’s penalty: resource arithmetic that model decisions only attenuate. What the benchmark measures here is *decision calibration, not capability*. Effect sizes, quota and squeeze parameters, rule-agent calibration, the zero-slack invariance argument, and the dose-response curves are in the supplementary material.

5.4 External Validity: Scorer Probed, Scaffold Measured

Is the scorer half general? We probed τ -bench’s official scorer (Yao et al. 2025) with zero model calls, replaying released trajectories through its released reward function and perturbing only the semantics: a verbatim passing trajectory scores 1.0, while a corrupted database write and a wrong answer, both shape-preserving, score 0.0. The scorer is outcome-based (it hashes the final database state), so the shape-scorer artifact is benchmark-specific (method and a trivial-agent finding in the supplementary material). The *scaffold* half, however, is present there as an uncontrolled axis, and we *measured* it: three models from two providers (Claude-Sonnet-4.6, Claude-Haiku-4.5, GPT-4o) across all four scaffolds the harness ships (tool-calling, few-shot, act, react; the LLM user simulator is Claude-Haiku-4.5 throughout, itself one of the evaluated agents, which we disclose). Changing nothing but the scaffold flag moves the same model’s reward by up to 0.267 and reorders the models (GPT-4o ties Sonnet under act yet finishes last under react), so a leaderboard that fixes any single scaffold can favor one model over another (Alzahrani et al. 2024). We report the slice as estimation only ($n=15/\text{cell}$, sign tests n.s., single-trial noise of the same order as the spread; full matrix in the supplementary material). A second probe, on BFCL’s AST scorer (Patil et al. 2025) (400 entries, five canned submissions each, offline), finds a third profile: the correct call passes in all 400 while value corruption, fabrication, wrong names, and empty arguments all fail (0 of 2,000 cells deviate), a *specification*-based scorer certifying spec validity rather than execution success.

Across the three benchmarks the scorer half is benchmark-specific, the scaffold half is uncontrolled wherever we probed it. Read as a two-factor design (four shipped scaffolds $\times$ three models, 15 tasks per cell), the same slice attributes $\eta^2 = 0.0293$ to the scaffold main effect against 0.0080 for the model. Both shares are small against the task-level residual, and the design is single-trial, so we read only their ratio and only as an ordering: on this benchmark’s own grid the measurement method accounts for more of the variance than the property being measured. At COMTRADEBENCH’s production configuration the corresponding model share is identically zero, the per-seed vectors being byte-identical.

5.5 The Validity Card

The audit’s deliverable is one card per benchmark; a *not probed* item is never read as passing. Two audits gate

the card’s *licensed claim*: whether the scorer is criterion-sensitive (it separates correct from degraded work product) and whether the model owns the execution-critical decisions the construct claims (Section 4.1). The strongest reading each configuration permits follows a fixed lattice: model capability only when both pass and the run replays; an outcome scorer over an uncontrolled scaffold, only model-scaffold system performance (τ -bench); a shape scorer, only self-report compliance (COMTRADEBENCH as shipped, repairable at L2); a specification scorer with scaffold not probed, only specification compliance (BFCL). The full per-item card is in the supplementary material (audits/validity_cards.md).

6 Limitations

B2 and B2-slack are exploratory. The pre-registered result is the B1 null; B2 and its slack ablation are the deferred enforced-quota variant ($n = 20/\text{arm}$, three models, one task family), offered for confirmatory follow-up. The zero-slack flip is resource coupling, not decision failure: once the quota binds, retry and skip are outcome-equivalent, so it cannot indict any model’s decisions, and we ran one slack level.

Single domain. COMTRADEBENCH is stylized trade-data-shaped extraction work from a seeded generator, and only ComtradeAdapter realizes the full ownership intervention (external targets are scorer-audited): cross-domain transfer is a design property, not yet empirical.

A small suite with a measured scale bound. The spectrum rests on five tasks $\times$ 10 seeds, two ceiling-saturated for the frontier. A sixth task (T7) is excluded by a solvability audit: its 750-row re-emission demand exceeds any single-submission budget (F_1 cap 0.26/0.46) and floors every model run on it, where L2 scoring stops measuring decisions and starts measuring output budgets (supplementary material).

Scope and provenance. B1 is powered only against large effects (power 0.998 at $n = 20$), so smaller escalation effects are not excluded; per-model scores are point-in-time on pinned API endpoints.

7 Conclusion and Future Work

When the scaffold owns the execution-critical decisions and the scorer grades shape, a benchmark score is not identified in the model: seven models submit byte-identical payloads, and a fabricated record set scores as high as the correct one. The joint repair recovers a spectrum in which the worst case reorders the mean, and the same confound changes the conclusion of our own pre-registered experiment. Four practices follow. *Measure the model, not the scaffold; score against ground truth, not shape; report worst-case, CVaR, and Reliability@ τ as first-class outputs; and treat the scaffolding level as a controlled variable*, stating whether adversity bills a quota. The protocol applies on top of existing benchmarks, not in place of them. Three directions remain open: a controlled ownership decomposition on a benchmark we do not own, which requires showing that reassigning decisions preserves task semantics; extending the audit beyond what a zero-call probe battery can reach, so that scorers requiring model calls to replay are assessed rather than recorded as *not*

probed; and protocol variants that lift the L2 interface floor so that de-scaffolded scoring extends to larger record sets.

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Supplementary Material

This supplement provides the extended analyses referenced in the main text and the complete experimental record. Unless noted otherwise, every table and figure is regenerated directly from the released result files by `gen_appendix.py`.

A Extended Results and Analysis

This section holds the full version of every passage the main text abbreviates. Nothing is dropped: all results, tables, and figures from the project appear here or in the appendices.

A.1 Pipeline and Scaffolding Spectrum (Figure)

Figure A1 gives the one-page visual summary of the measurement pipeline; the measured per-model scores at each scaffolding level (defined in the main text, Section 4.2) are in Figure A7.

A.2 B1: Instrument Deviation and Power Study (full)

The confirmatory B1 run used the substrate `T9_clean_ab` under the matched-mean A/B design, three models (Llama-3.3-70B, GPT-5, Claude-Sonnet-4.6), retries free. It deviated from the frozen pre-registration, and per the pre-registration’s own rule (any deviation is reported as exploratory) we disclose it: the frozen document specified the de-scaffolded slim agent with the terminal judge reward as the outcome, whereas the run used the L1 half-scaffold agent with ground-truth fetch coverage. The substituted instrument isolates the retry/skip decision and is therefore structurally blind to the pre-registered dedup/totals effect channel: coverage deduplicates fetched records into a set and filters totals rows before scoring, so duplicate injection and totals contamination cannot move the outcome at all, and free retries neutralize the 429 faults. The only channel through which a model could affect its score was an explicit SKIP. The pre-registered decision rule fires only if the paired bootstrap CI excludes 0 on the positive side *and* a sign-flip permutation test gives $p < 0.05$. The result is null for every model: $\Delta = +0.000$, 95% CI $[-0.007, +0.007]$, $p = 1.0$, paired Cliff’s $\delta = +0.05$, 95% CI $[-0.30, +0.40]$. The interval is tight against large effects only, and dominance beyond ± 0.4 is excluded. SKIP, meanwhile, was chosen in 0 of 120 episodes, so the one model-sensitive channel never fired at all. That is not a powered null, and we do not report it as one.

A Monte-Carlo power study (400 replications) shows the design detects a large effect ($\Delta = 0.209$, the magnitude of the prior “T9 gap”) with power 0.998 at $n = 20$, already 0.92 at $n = 12$, while holding size at 0.04–0.09. Medium ($\Delta = 0.058$) and small effects are not excluded. The simulation lets fragility act directly on the outcome, so the power figure certifies the statistical test and says nothing about whether the instrument could have picked up a behavioral effect. That prior gap (21 judge points, the $\Delta = 0.209$ above) was a confound (scaffolding plus a single seed plus a reasoning model running slow against a step budget), not a non-stationarity effect. Under free retries the always-retry policy makes both arms’ per-seed outcomes identical across

Model	Δ	95% CI	Cliff’s δ [CI]	p	p_{Holm}	Deaths	Skips
<i>B1: free retries (pre-registered), $n=20/\text{arm}$</i>							
All three models	+0.000	$[-0.007, 0.007]$	+0.05 $[-0.30, 0.40]$	1.00	n/a	n/a	0/0
<i>B2: enforced quota 8, squeeze -3 (exploratory)</i>							
Llama-3.3-70B	+0.087	$[0.051, 0.128]$	+0.55 $[0.25, 0.80]$	.0011	.0066	11/20	6/1
GPT-5	+0.063	$[0.026, 0.109]$	+0.25 $[-0.10, 0.60]$	.0076	.0380	8/20	9/4
Claude-Sonnet-4.6	+0.046	$[0.009, 0.091]$	+0.00 $[-0.40, 0.40]$	.0355	.1065	7/20	11/7
<i>B2-slack: quota 10, post-squeeze slack 2 (exploratory)</i>							
Llama-3.3-70B	+0.043	$[0.014, 0.075]$	+0.25 $[-0.10, 0.60]$	.0231	.0924	2/20	5/0
GPT-5	+0.019	$[-0.015, 0.053]$	+0.10 $[-0.25, 0.50]$	.3495	.6990	1/20	6/3
Claude-Sonnet-4.6	+0.010	$[-0.019, 0.040]$	+0.00 $[-0.35, 0.40]$	.6031	.6990	0/20	7/5

Table A1: Escalating vs. matched-mean constant stress. Δ = paired mean coverage difference (constant – escalating; positive = escalation penalty), with seeded paired-bootstrap 95% CI and sign-flip permutation p . Cliff’s δ = within-pair dominance $[\#(\Delta_i > 0) - \#(\Delta_i < 0)]/n$ with seeded bootstrap 95% CI: only Llama’s B2 penalty is a dominance effect; Claude’s nominally significant one is magnitude-driven ($\delta = 0.00$, equal wins and losses). p_{Holm} = Holm-adjusted p over the full six-test exploratory family (B2 + B2-slack; B1 is pre-registered and outside the family): only Llama’s and GPT-5’s B2 penalties survive correction. Deaths = escalating-arm quota exhaustions; Skips = SKIP actions, escalating / constant arm. B1’s rows are identical for all three models.

models within each arm, fixed by the seeded fault sequence rather than by anything a model did.

We additionally ran the pre-registration-faithful instrument on all three pre-registered models (slim agent, terminal judge reward, identical adversary; $n=20/\text{arm}$), and report it under the frozen decision rule without post-hoc adjustment. On the terminal judge reward Llama gives $\Delta = +0.005$ ($p = .386$, $\delta = +0.05$ $[-0.40, +0.45]$) and Claude $\Delta = +0.016$ ($p = .167$, $\delta = +0.35$ $[-0.05, +0.70]$). Both are null. GPT-5 gives $\Delta = +0.011$ with a bootstrap CI $[+0.002, +0.023]$ excluding zero, sign-flip $p = .034$, and $\delta = +0.25$ $[-0.15, +0.65]$. The pre-registration’s primary outcome was “ ≥ 1 frontier reasoning model,” so under its own rule this is a *confirmation*. The ground-truth scorer accounts for it: on the identical episodes GPT-5’s and Claude’s ground-truth F_1 deltas are -4×10^{-5} and its coverage delta is ≈ 0 (1.1×10^{-17}), and Llama’s deltas are null though noisier (coverage $\Delta = -0.047$, $p = .75$). The submitted data is equally correct across arms (identical unique-row, duplicate, and totals counts), so the pre-registered confirmation lives *entirely* in judge dimensions that ground truth does not measure. The escalation effect the pre-registration set out to find is real under the scorer it named and vanishes under ground truth: the scorer-dependence this paper diagnoses, surfacing inside our own confirmatory experiment (`results_b1_slim.json`).

A.3 B2 Rule-Agent Calibration (full table)

A.4 Zero-Slack Decision-Invariance and the Slack Regime (full)

What the flip does not show. At zero post-squeeze slack (quota 8 – 3 = 5 for 5 pages), retry and skip are outcome-equivalent (each fault costs one page either way), so once the

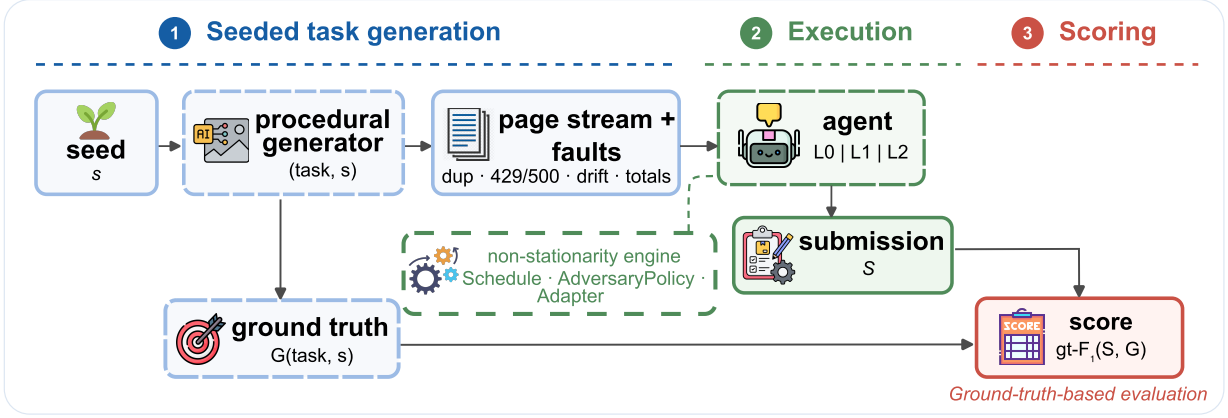


Figure A1: The COMTRADEBENCH pipeline in three stages. (1) *Seeded task generation* (blue): a seed drives the procedural generator, which emits the canonical ground-truth record set G and serves a fault-injected page stream (duplicates, 429/500, page drift, totals rows); (2) *execution* (green): the agent, at scaffolding level L0–L2, produces a submission S , with the non-stationarity engine (Schedule, AdversaryPolicy, EnvironmentAdapter) at the agent–environment boundary; (3) *scoring* (red): S is scored as ground-truth F_1 against G . The measured per-model scores at each scaffolding level are in Figure A7.

Rule policy	Δ	95% CI	p	Skips	Deaths
<i>Zero slack (quota 8, squeeze −3)</i>					
always-retry	+0.096	[0.061, 0.137]	.0004	0	12
quota-aware	+0.063	[0.026, 0.109]	.0076	12	9
always-skip	−0.014	[−0.055, 0.027]	.5726	34	4
<i>Slack 2 (quota 10, squeeze −3)</i>					
always-retry	+0.015	[−0.003, 0.039]	.3017	0	2
quota-aware	+0.027	[−0.009, 0.063]	.2062	9	2
always-skip	−0.014	[−0.055, 0.027]	.5726	35	0

Table A2: No-LLM rule-agent calibration on the identical B2 protocol ($n=20$ paired seeds; skips and quota deaths are totals over both arms’ 40 episodes). At zero slack the fixed always-retry policy reproduces the flip ($\Delta = +0.096$, exceeding every model in Table A1), and the deterministic quota-aware rule is bit-identical to GPT-5’s summary; at slack 2 no rule policy is significant.

squeeze lands, the escalating arm is decision-invariant by construction. The data confirms it: the escalating-arm per-seed coverage vectors are *identical* for Llama and GPT-5, and Claude’s differs on exactly 1 of 20 seeds (all vectors are in the released JSON). The escalation penalty in this configuration is therefore *resource arithmetic, not decision failure*: no high-coverage policy could have avoided it (the always-skip rule avoids it in the Δ sense, $\Delta = -0.014$, n.s., but only by depressing coverage in both arms, 0.704/0.690 against the 0.83 ceiling).

Where slack exists, models differ in a counterintuitive direction. The constant arm never triggers the squeeze and keeps 3 pages of slack, so retrying is affordable there; and that is where the models come apart. Llama plays it near-optimally: 1 skip in 20 episodes, coverage 0.823, near the 0.83 duplicate-overwrite ceiling. Claude skips 7 times *unnec-*

essarily and pays for it (0.774): the scarcity framing alone, with the budget never actually at risk (0 constant-arm quota deaths), induces over-conservatism in the frontier model. The slack ablation (quota 10, so 2 pages of post-squeeze slack) flips the value of that same trait. Claude’s skip-hedging now buys insurance: zero quota deaths, and the escalation penalty vanishes ($\Delta = +0.010$, n.s.). GPT-5 likewise escapes it ($\Delta = +0.019$, CI $[-0.015, +0.053]$, $p = .3495$, n.s., one quota death). Llama keeps a nominally significant penalty ($\Delta = +0.043$, uncorrected $p = .0231$; $p_{\text{Holm}} = .0924$, 2 quota deaths). The rule calibration re-attributes that persistence: at slack 2 the always-retry rule is *not* significant ($\Delta = +0.015$, $p = .30$; Table A2), so retrying alone no longer carries a penalty. Llama’s constant arm matches the always-retry rule exactly (0.832), while its escalating arm (0.789 against the rule’s 0.817) is degraded by its five chosen SKIPS; the persistent penalty is skip-induced over-conservatism under escalating scarcity, not retry-heaviness. Skip propensity is thus a cost under abundance and a hedge under escalating scarcity, and only its timing decides which. Neither policy dominates.

Dose response (full curve). Two further budget configurations (quota 9 and 12, i.e. post-squeeze slack 1 and 4; $n=20/\text{arm}$ each) complete a dose-response curve over slack (Figure A3). We report these as *estimation*, not additional confirmatory tests. The penalty decays with slack for all three models, monotonically for Llama $+0.087 \rightarrow +0.074 \rightarrow +0.043 \rightarrow +0.008$ (slack 0 $\rightarrow 1 \rightarrow 2 \rightarrow 4$) and GPT-5 $+0.063 \rightarrow +0.033 \rightarrow +0.019 \rightarrow +0.008$, and to a negligible level for Claude $+0.046 \rightarrow +0.041 \rightarrow +0.010 \rightarrow +0.013$ (whose slack-2 and slack-4 points are statistically indistinguishable). Llama’s CI excludes zero through slack 2; GPT-5’s and Claude’s include zero from slack 2 onward. At slack 4 the penalty is negligible for every model; with four spare requests nothing binds, so decisions are once more

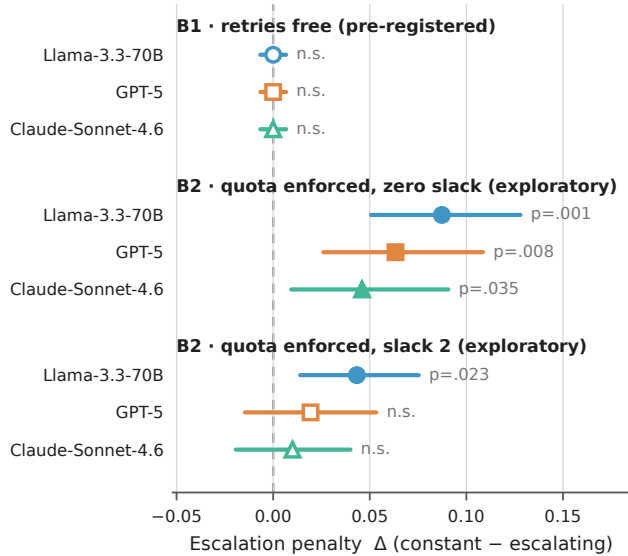


Figure A2: The escalation penalty Δ (positive = escalating trajectory hurts beyond matched-mean constant stress) with 95% bootstrap CIs, $n=20$ paired seeds per arm. Under free retries (B1, pre-registered) the penalty is zero for all three models. Adding one mechanism (an enforced request quota that every fetch consumes, squeezed mid-episode only in the escalating arm) makes the penalty positive on all three uncorrected (B2); after Holm correction it survives for Llama and GPT-5 only, and a fixed no-LLM always-retry policy achieves $\Delta = +0.096$, above every model. With post-squeeze slack, the penalty persists only for Llama and vanishes for the skip-hedging Claude and GPT-5. Filled markers: CI excludes 0; hollow: n.s. All escalation experiments (B1, B2, the slack and dose ablations, and the degradation curves) use exactly the three models named in the frozen pre-registration; Claude-Fable-5 postdates the freeze and is absent from the B-series by design, not omission.

outcome-free. Decision relevance is thus an *inverted U* in slack: at zero slack every choice is forced, at high slack no choice matters, and models can be told apart only in the regime between.

A.5 Frontier Within-Model Fragility and the T8 Mixed-Fault Case

The reliability metrics are also a within-model discipline for the frontier tier. GPT-5 and Claude-Sonnet score $F_1 = 1.0$ on dedup and both retry tasks but drop to 0.80 on page-drift, a recall failure (recall 0.67): the models emit clean, deduplicated rows but miss a third of the canonical set when page contents are non-deterministic (Claude-Fable-5 clears page-drift at 1.000). The mixed-fault task T8, now complete for all eight models and part of main-text Figure 2, separates the frontier further. GPT-5’s per-seed record is bimodal at the top, [1, 1, 1, 1, 0, .81, 1, 1, .81, .81]: six perfect episodes, three at 0.81, and one full-episode zero (mean 0.844, worst 0.000), handing the 3rd-ranked model by mean the suite’s

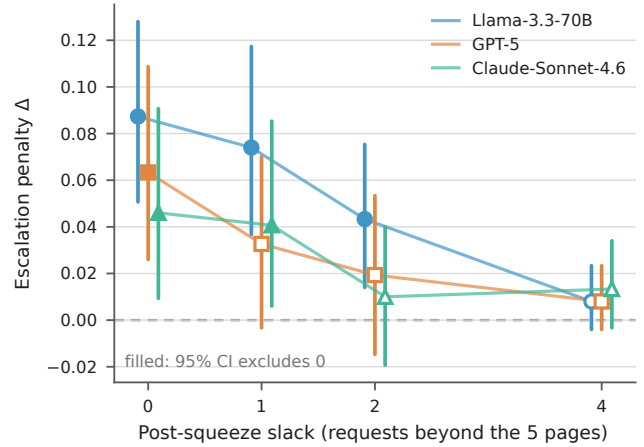


Figure A3: Dose response: escalation penalty Δ (95% bootstrap CI) versus post-squeeze slack, all three models at $n=20$ /arm per point (the pre-registered trio; Claude-Fable-5 postdates the freeze and is absent from the B-series by design). Filled markers: CI excludes 0. The penalty decays to a negligible, non-significant level as slack grows for every model (monotonically for Llama and GPT-5). Reported as estimation; the confirmatory family is unchanged.

largest reliability gap (0.930). Claude-Fable-5 and Claude-Sonnet-4.6 show the complementary profile, never a perfect T8 episode (means 0.869/0.914) but never a collapse (worst 0.824/0.824), while Claude-Haiku-4.5, GPT-4o, and GPT-4o-mini sit at a flat 0.824 (recall 0.70, precision 1.00, near-deterministic across seeds). Consistent but capped and sometimes perfect but brittle are distinct reliability profiles at a similar mean.

A.6 T7: the L2 Re-Emission Ceiling, Measured

T7 (totals trap) scales the totals-row fault family to 750 canonical rows served at page size 250. A solvability audit shows that under the L2 protocol this cell is not a fair reliability measurement. The slim agent must re-emit every cleaned row through a single `submit_results` call, and one row costs ≈ 54 tokens of output. The budgets we grant (6,000 tokens non-reasoning, 12,000 reasoning) therefore cap recall at $\approx 0.15/0.30$ and F_1 at $\approx 0.26/0.46$ regardless of model ability. The measured outcomes sit far below even those caps and collapse for every model run: GPT-4o-mini 0.000, GPT-5 0.017, Claude-Haiku 0.023, GPT-4o 0.026, Claude-Fable-5 0.043, Claude-Sonnet 0.127 (all $n=10$; per-seed vectors in Table A13), with a characteristic high-precision/near-zero-recall signature (precision ≈ 1.0). We therefore report T7 as a *measured scale bound of de-scaffolded scoring* rather than as a Table-1 cell. It marks the point where the L2 re-emission demand stops measuring execution decisions and starts measuring output budgets. It is the mechanical-floor channel that floors Llama at small N , now binding every model at large N . Protocol variants that lift the bound (chunked submission, id-only re-emission) are deferred to future work; Llama and

Qwen were not run on T7 (their L2 floor at $N \leq 60$ already binds).

A.7 Model Access and Decoding Disclosure

All E1 episodes call provider APIs through one OpenAI-compatible client against pinned endpoints: `api.anthropic.com/v1` (model ids `claude-fable-5`, `claude-sonnet-4-6`, `claude-haiku-4-5`), `api.openai.com/v1` (`gpt-5`, `gpt-4o`, `gpt-4o-mini`), and `api.together.xyz/v1` (`Llama-3.3-70B-Instruct-Turbo`, `Qwen2.5-7B-Instruct-Turbo`); rows were collected June–July 2026 (the T7/T8 completion runs on 2026-07-08 through 2026-07-10). The client requests `temperature=0` and an output budget of 6,000 tokens (12,000 for the reasoning-class Claude-Fable-5, GPT-5, and Claude-Sonnet-4.6). Two provider quirks, pinned by tests in the released client, qualify that setting: GPT-5 rejects the `temperature` and `stop` parameters (and takes `max_completion_tokens`), and Claude-Fable-5’s endpoint rejects `temperature` as deprecated for that model, so those two models sample at their providers’ defaults rather than at a client-pinned 0, and their per-seed variance (Table A13) should be read with that in mind. All other models run at `temperature=0`. Per-model scores are point-in-time measurements of these dated endpoints, as the main-text Limitations states.

A.8 GPT-4o Bimodality: Full Mechanism

The worst-case reorder rests on GPT-4o’s genuine bimodality on dedup (T3): per-seed $F_1 = [0, 0, 1, 0, 0, 0, 0, 1, 1, 1]$ at $n=10$, extended to 14/30 solved at $n=30$ (Wilson 95% CI $[0.30, 0.64]$; Appendix C.6). Every episode opens with the same off-by-one call, `fetch_page(page=0)` against the 1-indexed API, deterministically rejected with HTTP 422. All 14 solves recover by probing page 1 after three or four rejections. The 16 failures re-issue the byte-identical call four to nine times; the 422 carries the same `retry: true` flag as a genuine transient fault, so the repeated call is consistent with the retry signal the model receives. Each then *knowingly* submits a schema-valid payload with zero data rows and a run log stating the data could not be fetched (recall = precision = 0.00). The failures thus end in the same empty-payload channel that floors Llama and Qwen, entered upstream: rows never fetched, rather than fetched and not re-emitted. The outcome is fully determined by whether the model ever varies the page argument (14/14 of solves, 0/16 of failures): the bimodality is identifiable error-recovery behavior in the transcript, not mysterious seed-to-seed variance. What is model-dependent is the *frequency* of entering that channel under an identical demand (GPT-4o escapes on 14 of 30 runs, the floored tier on none), so the worst-case reorder rests on error-recovery reliability, not on the dedup decisions themselves: every episode that obtained rows deduplicated them perfectly. GPT-4o-mini shows the identical terminal signature one tier down (bimodal on page-drift).

A.9 External Scorer Probe: Method and a Trivial-Agent Finding

We probed the official scorer of τ -bench (Yao et al. 2025) with zero model calls, replaying its released historical trajectories through its released reward function (`external/taubench_probe.py`) and perturbing only the semantics. A verbatim passing trajectory scores 1.0; the same trajectory with one database write semantically corrupted (shape preserved) scores 0.0; a trajectory whose required answer string is wrong (shape preserved) scores 0.0. τ -bench’s scorer is outcome-based (it hashes the final database state) and therefore does *not* carry the shape-scorer artifact: the confound’s two halves are separable design properties, not a universal condition. The *scaffold* half, however, is present as an uncontrolled axis: the released harness ships four selectable scaffolds whose differences are material and silent: the tool-calling scaffold truncates parallel tool calls to the first and treats malformed tool-call JSON as episode-fatal, while the ReAct scaffold forgives the same error. The probe additionally surfaces a trivial-agent artifact even in this outcome-based benchmark: 5 of 115 τ -bench retail test tasks are satisfied by an empty “do-nothing” trajectory, because their ground truth requires no database write and no output, an outcome-based analogue of the empty-submission floor we document on our own judge.

A.10 External Scaffold-Variance Slice: Full Matrix

We measured scaffold variance on τ -bench directly: three models from two providers (Claude-Sonnet-4.6, Claude-Haiku-4.5, GPT-4o) $\times$ all four scaffolds the harness ships, 15 retail test tasks per cell (task ids 0–14, one trial), with the LLM user simulator held fixed (Claude-Haiku-4.5) and `temperature=0`. No new code: only the `--agent-strategy` flag changes between cells.

Scaffold	Sonnet-4.6	Haiku-4.5	GPT-4o
tool-calling	0.733	0.667	0.667
few-shot	0.733	0.533	0.600
act	0.667	0.600	0.667
react	0.533	0.467	0.400

Table A3: τ -bench retail, average reward over the same 15 tasks per cell. Changing only the scaffold flag moves the same model by up to 0.267 (GPT-4o); react is the worst scaffold for all three models; and few-shot triples the Sonnet–Haiku gap. The slice measures scaffold variance *within* a model, not model ranking; the user simulator is Claude-Haiku-4.5 throughout, itself one of the evaluated agents (disclosed).

Four observations. (1) *Absolute scores do not transfer across scaffolds*: Sonnet spans 0.733–0.533, Haiku 0.667–0.467, and GPT-4o 0.667–0.400, a swing of up to 0.267 (4 of 15 tasks) for the identical model, tasks, and user; react is the worst scaffold for all three models. (2) *Ordering is scaffold-dependent in general*: across tool-calling, act, and react the Sonnet–Haiku contrast is monotone and rank-preserving (Sonnet ahead by exactly one task at each level),

but few-shot is Sonnet’s joint-best scaffold and Haiku’s second-worst (a scaffold–model interaction that triples that gap), and GPT-4o crosses between the two: it ties Haiku under tool-calling, ties Sonnet under act, sits between them under few-shot, and finishes last under react. The apparent ranking therefore depends on which scaffold the leaderboard fixes. (3) *Statistical footing*: per-model paired sign tests between the extreme scaffolds do not individually reach significance at $n=15$ (Sonnet 4 up/1 down, $p = .375$; Haiku 5 up/2 down, $p = .453$; GPT-4o 6 up/2 down, $p = .289$); the slice is reported as estimation, with its strength in the cross-cell consistency rather than any single contrast. (4) *Test–retest*: a full rerun of the (Sonnet, tool-calling) cell under the identical configuration scores 0.867 against the original 0.733. Thirteen of 15 tasks agree, and both flips, tasks 4 and 14, are $0 \rightarrow 1$. Single-trial noise on this cell (0.133) is therefore of the same order as the cross-scaffold spread, which is why we report the slice as estimation rather than as a hypothesis test. Raw result files, per-task pairings, the rerun (retest_tool-calling-sonnet-4-6_0709155107.json), and the exact run commands are in `external/taubench_slice/`. Figure A4 plots the matrix as an interaction chart, showing the within-model spread, the react floor, and the few-shot split.

One caveat on the few-shot column. Kapoor et al. (2026) report that τ -bench’s official few-shot agent loads demonstration data that overlaps the benchmark’s own test set, and exclude that scaffold from their analysis; the file they identify is in the airline domain, while our slice is retail. The headline swing of 0.267 (GPT-4o, tool-calling versus react) does not pass through few-shot and is unaffected. We record the issue because the few-shot cells, and the Sonnet–Haiku gap we read off them, should be interpreted with it in mind.

A.11 The Non-Stationarity Engine (full)

The main text summarizes the engine; here is the full architecture. To generate stress reproducibly, and to test whether its *shape* matters, a domain-agnostic engine decouples three concerns. `Schedule` maps an `AgentProgress` snapshot to an intensity $i \in [0,1]$: the shape of the non-stationarity. Intensity is driven by the agent’s *success* count, not elapsed steps (linear, step, and plateau schedules). `AdversaryPolicy` maps progress to a `Perturbation`, an *intent*: a transient fault, an observation corruption (duplicate/reorder/noise/drop), a budget delta, or a latency. The reference `EscalatingAdversary` sets fault probability to $\text{base} + \text{scale} \cdot i$ and corruption rate to $\text{scale} \cdot i$; it knows nothing about trade data. `EnvironmentAdapter`, the only domain-coupled piece, realizes an intent in one environment: `ComtradeAdapter` produces 429 responses, injected duplicate rows, and a totals-row distractor once intensity crosses `totals_at`. The `AdversarialWrapper` ties them at the agent–environment boundary, appending per-step intensity to a trace the metrics consume; a fault cancels a success, so escalation tracks real progress. Reproducibility is built into the framework: per-step randomness is `Random(sha256("{seed}:{step}"))`, so the same seed yields a bit-for-bit identical adversarial sequence.

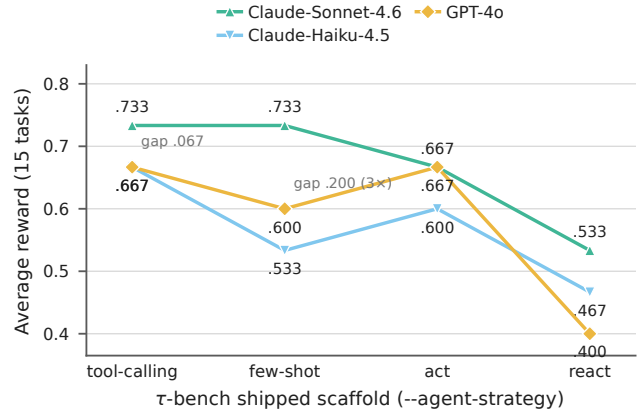


Figure A4: The τ -bench scaffold–model interaction: Table A3 as lines. Each line is one agent across the four shipped scaffolds (same 15 tasks, fixed user simulator); react is the worst scaffold for all three models, the within-model spread reaches 0.267 (GPT-4o), and few-shot triples the Sonnet–Haiku gap (0.067 $\rightarrow$ 0.200) while GPT-4o crosses between the two: the induced ranking is scaffold-dependent. Recomputed from the released per-episode JSONs by `Figures/figure8_interaction.py`.

Controlled stress for Claim B. The engine supplies the matched-mean A/B design the main text pre-registers. The substrate is a clean task (75 rows over 5 pages, own fault mode disabled), so the wrapper is the sole source of adversity and the arms differ only in difficulty *trajectory*: the escalating arm ramps $0 \rightarrow 1.0$; the constant arm holds the ramp’s mean, 0.60. Both apply 429 faults and duplicate injection at matched mean, and both inject totals-row contamination once intensity ≥ 0.70 , a threshold the escalating arm crosses late and the constant arm never reaches: the nonlinearity through which the pre-registration expected the two matched-mean arms to diverge. The confirmatory run’s coverage outcome neutralizes this channel by construction, a deviation disclosed and analyzed with the B1 results above.

A.12 Agent-Layer Evaluations (full)

The three optional LLM-assisted layers of the pipeline are measured, not assumed. A scaffold-discovery step that proposes ownership maps from harness source classifies 13 to 14 of 15 decisions correctly in each of three independent runs against this paper’s three declared reference harnesses (the recurrent miss is the one genuinely ambiguous cell). A synthesis step that generates a de-scaffolded harness variant passes compilation, an independent discovery cross-audit (all five decisions model-owned), and a live ground-truth-scored episode. A probe-design step, given only a scorer’s source, predicts the scorer kind for all three audited scorers and independently reproduces the human probe batteries at 3 of 4 to 4 of 4 coverage under a fixed keyword rubric, and adds probes the human batteries lacked. A narration layer is constrained by a number guard: any figure in generated prose that does not appear in the audit JSON is rejected.

A.13 The Full Validity Cards

Table A4 is the complete per-benchmark validity card; the main text summarizes the licensed-claim lattice. Every cell is pass, fail, or *not probed*, with the evidence inline. The columns disagree in different directions: the audit discriminates conditions rather than finding the same defect everywhere.

Decision rules. Each row is populated deterministically from the audit artifacts under a fixed rule. *Criterion sensitivity* passes iff the scorer strictly separates ground-truth quality on the forged battery, i.e. $j(\text{correct}) > j(x) + \varepsilon$ for every submission x of strictly lower ground-truth quality (empty, fabricated, duplicate-laden), with ε the scorer’s resolution; it fails on any tie or inversion across a material quality gap (COMTRADEBENCH: $j(\text{fabricated}) = j(\text{correct})$, gap 0) and is *not probed* when no canonical-correctness oracle can be constructed. *Ownership completeness* passes iff $D_{\text{claimed}} \subseteq D_{\text{model}}$ and every claimed decision is non-degenerate ($\text{Var}(A) > 0$ with outcome sensitivity); it fails when a claimed execution-critical decision is harness-owned or degenerate (τ -bench: reward moves 0.267 across shipped scaffolds; COMTRADEBENCH L1: SKIP 0/120) and is *not probed* when the harness is not inspectable (BFCL: single-turn decoding scaffold). *Replay* passes iff runs reproduce bit-for-bit from (task, seed). The *licensed claim* is the output row, and it follows a fixed lattice:

- **model capability** when replay, criterion sensitivity, and ownership completeness all pass;
- **model-scaffold system performance** when criterion passes but ownership fails;
- **self-report compliance** when the scorer is shape-based, so criterion fails;
- **specification compliance** when a specification scorer passes and ownership is not probed.

A not-probed row never licenses the corresponding claim.

Audit coverage and cost. All three targets are audited by one key-free pipeline that makes *zero* model calls: it regenerates every COMTRADEBENCH card cell from the seeded core, replays τ -bench’s released reward function and scaffold grid, and runs BFCL’s vendored AST checker offline. Table A5 reports what each target exercises. Integrating a target costs benchmark-specific code (COMTRADEBENCH: ComtradeAdapter, 111 lines; τ -bench: 196; BFCL: 174) against a shared, benchmark-agnostic core (probe battery, joint intervention, reliability, and card synthesis; ≈ 1000 lines). The complete run finishes in seconds, and its 13 built-in oracle checks (6 for COMTRADEBENCH, 4 for τ -bench, 3 for BFCL) all pass, so the auditor is itself an executable regression suite. τ -bench and BFCL are integrated as benchmark-specific audit modules feeding the shared card synthesizer, not full re-implementations of the ownership intervention: τ -bench’s ownership evidence is scaffold variance (not a controlled de-scaffolding) and BFCL’s scaffold is not probed. A controlled ownership decomposition on τ -bench is future work.

Target interface. A benchmark is audited through four key-free hooks, which ComtradeAdapter implements for COMTRADEBENCH:

- a *scaffold-ownership map*, assigning each execution-critical decision (retry, dedup, totals-drop, pagination, submission) to harness or model at each level. It is declared in v0, and agentially discovered then human-confirmed in v1;
- a *task registry* enumerating the seeded tasks;
- a *canonical-truth oracle* returning ground truth from (task, seed);
- a *scorer hook* that replays the benchmark’s own scorer on a submission.

Table A6 shows how the two external targets supply these hooks.

Failure modes. When a hook is unavailable or a model-owned channel is degenerate, the audit does not fabricate a verdict; it records *not probed* or *fail* by the Decision-rules above. Table A7 enumerates the resolutions.

B Extended External Audit: a Scorer-Kind Census Across Nine Benchmarks

The main text audits three benchmarks (Section 5.4). This section widens that audit to nine, to ask how far the two halves of the confound generalize. Everything here is offline and makes zero model calls: scorer classifications are grounded in each benchmark’s released scorer source, and executed deviation counts are reported only where the scorer was actually run on a canned battery; source-level classifications are labeled as such.

We classify each shipped scorer into one of four kinds. *Shape / self-report*: grades output format, field presence, or self-reported metadata, never the submitted values against ground truth (a well-formed fabrication can score as high as the truth). *Outcome*: grades the final answer or world state against ground truth (test pass/fail, final DB state, answer match, produced-file content). *Specification*: grades conformance to a required spec (e.g. AST match), certifying validity rather than execution success. *LLM-judge*: an LLM decides pass/fail or win/lose; it grades *plausibility* rather than ground truth (in the audited case no gold answer exists at all), so a confident, well-formed fabrication can be labeled as passing, and every verdict requires a model call, so this kind is *not* offline-probeable.

Three observations follow. First, the *shape* defect of Section 3 (main text) is benchmark-specific, and the census quantifies this: exactly one of nine benchmarks (COMTRADEBENCH, as shipped) carries a pure shape / self-report scorer, and most shipped scorers are value-grounded. As an executed spot-check of one of them, we ran GAIA’s released `question_scorer` (vendored, unmodified) on a canned battery over five gold-answer types: correct submissions pass 5/5, formatting variants (whitespace, case, \$/% units) pass 5/5, and fabricated-but-well-formed values pass 0/5.

Audit item	COMTRADEBENCH (as shipped)	τ -bench (retail slice)	BFCL (AST slice)
Model-sensitive ownership	fail : L0 owns 5/5 decisions; submissions byte-identical, 21/21 model pairs (7 models, 3 providers)	fail : reward moves 0.267 across the 4 shipped scaffolds; ranking scaffold-dependent	not probed
Criterion-aligned scorer	fail : fabricated = correct (0.980), gt F_1 0 vs 1	pass : corrupt DB write $\rightarrow$ 0.0 (aligned to final DB state)	pass : spec-based, 0/2000 deviate
Null floor	fail : empty submission scores 0.645	pass : no shape floor observed	pass : empty arguments fail
Trace preference	fail : misreported correct data -0.398	not probed	not probed (no trace channel)
Decision non-degeneracy	fail : L1 SKIP 0/120 (always-retry)	not probed	not probed
Interface floor	fail : L2 re-emission floor (floored tier)	not probed	not probed
Joint intervention	separation only at $L2 \times gt$ (main-text Table 1)	not probed (scorer swap unnecessary)	not probed
Reliability reporting	pass : 8 models, 50–70 seeded episodes each	est. only: 15 distinct tasks/cell, one trial each (not seeded repeats)	not probed (scorer-only audit)
Valid interpretation	L0 = system; L2+gt = model execution within measured interface bounds	per model–scaffold configuration	spec-valid calls
Invalid interpretation	shipped leaderboard as model capability	single-scaffold ranking as model ranking	spec pass = execution success
Overall verdict	repair required; model-valid after repair, within measured interface bounds	system-valid, scaffold-sensitive	scorer spec-valid; scaffold unaudited

Table A4: Full validity cards for the three audited benchmarks (numbers from the released BenchAudit artifacts). *Not probed* is a first-class cell value: an unaudited item is never reported as passing. The judge-probe cells come from the audit run on the 25-row dedup task; Table A10 puts the same three canned submissions through the same scorer on T2_multi_page (2,345 rows) and reports 0.987 and 0.648. The two runs differ only in the observability dimension, and the fabricated-equals-correct tie holds at both sizes.

Target	Stages audited	Oracle checks	Model calls
COMTRADEBENCH	scorer probe, 2×2 joint intervention, seeded reliability, ownership map, card	6/6	0
τ -bench (retail)	scorer-probe replay, scaffold variance	4/4	0
BFCL (AST slice)	scorer probe (AST checker)	3/3	0

Table A5: BenchAudit cross-target coverage from the key-free audit run (zero model calls; each stage completes in under a second). *Oracle checks* are the pipeline’s built-in regression assertions; all pass, reproducing every reported number.

Hook	Consumed by	COMTRADEBENCH	τ -bench	BFCL
Ownership map	ownership audit, 2×2	declared L0/L1/L2	4 shipped scaffolds (uncontrolled)	not probed
Task registry	all stages	seeded generator	retail slice	AST slice
Canonical-truth oracle	ground-truth scorer, criterion probe	seeded generator	final DB state	annotated spec
Scorer hook	criterion probe, 2×2	shipped judge	released reward fn	AST checker

Table A6: The four target hooks and how each audited benchmark supplies them. A missing hook yields *not probed* for the stages that consume it, never a pass.

Second, the census surfaces a fourth scorer kind beyond the three named in the main text: *llm-judge* (ToolBench’s Tool-Eval as the primary case; WebArena’s fuzzy-match branch and one AgentBench environment as minor cases). Shape scoring and llm-judging are two routes to the same criterion-validity failure: neither anchors the score to ground truth, so

a well-formed fabrication can pass. The main text diagnoses the first. The census adds the second, which zero-call probes cannot reach at all.

Third, the census explains why the main text’s 2×2 joint intervention (Table 1, main text) is difficult to replicate externally. That design needs a shape scorer *and* a constructible

Condition	Audit response
No canonical-truth oracle	criterion probe = not probed (metamorphic tests only)
No scorer access	scorer rows = not probed
Harness not inspectable	ownership = not probed (BFCL)
Ownership map declared but unverified	flagged for human confirmation (stage 1)
Non-deterministic replay	replay/reliability = fail
Model-owned decision degenerate (Var = 0)	ownership = fail (COMTRADEBENCH L1: SKIP 0/120)
De-scaffolding alters task semantics	intervention not licensed (equivalence unestablished)

Table A7: Failure modes and the audit’s deterministic response. *Not probed* and *fail* are first-class outcomes; an unmet condition is never reported as a pass.

Benchmark	Primary scorer kind	Fabricated = correct?	Offline-probeable?	Evidence
COMTRADEBENCH	shape / self-report	yes	yes	executed
τ -bench (Yao et al. 2025)	outcome (DB state)	no	yes	executed
BFCL (Patil et al. 2025)	specification (AST)	no (0/2000)	yes	executed
GAIA (Mialon et al. 2024)	outcome (answer)	no (0/5)	yes	executed
SWE-bench (Jimenez et al. 2024)	outcome (tests)	no	partial	source
WebArena (Zhou et al. 2024)	outcome + llm-judge	no (partial)	partial	source
AgentBench (Liu et al. 2024)	mostly outcome	mostly no	partial	source
OSWorld (Xie et al. 2024)	outcome (state)	partial	partial	source
ToolBench (Qin et al. 2024)	llm-judge	yes	no	source

Table A8: Scorer-kind census. *Fabricated = correct*: can a fabricated but well-formed submission score as high as a correct one? *Offline-probeable*: can the scorer run as a pure function on canned submissions with zero model calls (“partial”: the score math is pure but acquiring the end state needs a live environment). *Evidence*: “executed” = a canned battery was run through the released scorer; “source” = classification from the released scorer code, decisive lines inspected, deviation counts not run. Distribution of primary kinds: shape 1, outcome 6, specification 1, llm-judge 1.

ground truth in the same benchmark, and the two rarely co-occur: value-grounded benchmarks lack the broken column, while llm-judge benchmarks, which run against live services with no gold answer, lack the truth column. COMTRADEBENCH’s seeded generator supplies both, so it can serve as a controlled testbed for the repair and not only as a diagnosis.

C Complete Experimental Record

This appendix carries the complete experimental record behind the aggregates in the main text. Every data-bearing table below is generated directly from the released result files or released source code by `gen_appendix.py`; nothing is transcribed by hand. The one exception is the qualitative positioning matrix (Table A9), which contains no measured numbers.

C.1 Positioning Matrix

The qualitative matrix promised in the related-work section. Cells are our reading of each suite’s released evaluation protocol, and “partial” entries are marked deliberately in the prior suites’ favor. τ -bench’s pass^k does report cross-trial reliability, though without seeded worst-case or CVaR estimation over environment adversity. SWE-bench admits multiple community scaffolds, so multi-scaffold comparisons exist, but the scaffolding level is never a controlled experimental variable. Harness-Bench varies complete harness configurations factorially across model backends and explicitly dis-

claims mechanism-level decomposition, with one trajectory per task–model–harness cell.

C.2 Judge Probe: Three Canned Submissions Through the Released Scorer

The direct verification of the shape-scoring claim (Section 3 (main text)). `probe_judge.py` passes three canned submissions for `T2_multi_page` through the *released* `judge.py`, unmodified: (a) an empty `data.jsonl` with well-formed, self-consistent metadata and a disciplined run log; (b) 2,345 fabricated rows with the correct schema, plausible values, self-consistent metadata, and invented record ids (zero overlap with the seeded ground truth); (c) the exact seeded ground-truth record set, emitted by the benchmark’s own generator. All three cases carry the *identical* run log and metadata shape; only the submitted record set (and the truthful declared `row_count`) varies. The fabricated and correct submissions receive the same judge total and the same six-dimension breakdown; only ground-truth F_1 separates them.

C.3 Per-Task Parameters

The complete parameter set of the ten benchmark tasks plus the clean A/B base task, read directly from the released `server/tasks.py`. Every page of data is a deterministic function of `(task_id, seed)`; the fault column lists the injection mode with its mode-specific parameters.

Suite	Adversity location	Worst-case / cross-seed reliability	Scaffolding level as controlled variable
ToolBench (Qin et al. 2024)	prompt/task specification	no (mean pass rate)	no
BFCL (Yan et al. 2024)	none (static calls)	no (mean accuracy)	no
API-Bank (Li et al. 2023)	none (static dialogues)	no (mean accuracy)	no
AgentBench (Liu et al. 2024)	task environments, not fault injection	no (mean success)	no
τ -bench (Yao et al. 2025)	rule/policy level	<i>partial</i> : pass ^k over repeated trials; no seeded worst-case or CVaR	no
SWE-bench (Jimenez et al. 2024)	none (static issues)	no (mean resolve rate)	<i>partial</i> : community scaffolds vary (Yang et al. 2024; Xia et al. 2025), but the level is not an experimental axis
Harness-Bench (Yao et al. 2026)	none (sandboxed offline workflow tasks)	no (one trajectory per task-model-harness cell)	<i>partial</i> : six complete harnesses varied factorially across eight backends with native behavior preserved; configuration-level diagnosis, not an ownership axis
COMTRADEBENCH	environment response dynamics (seeded fault injection)	worst-case, CVaR@0.2, Reliability@ τ over seeded adversary draws	yes: ordinal L0/L1/L2 ownership axis

Table A9: Positioning matrix over the seven suites cited in the related-work section. Qualitative; no measured numbers.

Submission	Rows	Corr./30	Compl./15	Rob./15	Eff./15	DQ/15	Obs./10	Judge total	ground-truth F_1
empty	0	18.00	15.00	15.00	7.50	5.00	4.33	0.648	0.000
fabricated	2345	30.00	15.00	15.00	15.00	15.00	8.67	0.987	0.000
correct	2345	30.00	15.00	15.00	15.00	15.00	8.67	0.987	1.000

Table A10: Judge probe on T2_multi_page: six-dimension breakdown (points), judge total (/100, normalized to $[0, 1]$), and ground-truth F_1 of the same submission, from probe_judge_results.json. The empty case trips the correctness governance gate ($< 70\%$), which halves its efficiency and observability credit; it still floors at 0.648. The fabricated and correct record sets are indistinguishable to the judge (0.987 each) and maximally separated by ground-truth F_1 (0.0 vs. 1.0).

C.4 E1: Complete Per-Task Statistics, Including Diagnostic Runs

The main-text Figure 2 scopes aggregates to the five fully run execution tasks (T3–T6, T8), $n=10$ seeds per cell for all eight models. The complete record below additionally includes the T7 re-emission-ceiling diagnostic (Section A.6; excluded from aggregates because its 750-row re-emission demand exceeds single-submission output budgets, capping F_1 at 0.26/0.46 for non-reasoning/reasoning token limits) and the partial T10 runs, with the per-cell seed count n disclosed. Metric: ground-truth F_1 per episode; CVaR at $\alpha=0.2$; Reliability at $\tau=0.5$.

C.5 E1: Raw Per-Seed F_1 Vectors

The seed-level grid behind Table A12. Solve-or-fail bimodality (e.g. GPT-4o on T3) is directly visible.

C.6 Failure Forensics: GPT-4o on T3 Dedup at $n=30$

The T3 bimodality of the main text was re-run at $n=30$ with the full message transcript, submitted payload, and judge breakdown captured per episode (harness_gpt4o_forensics.py

→ results_gpt4o_t3_n30.json and forensics_gpt4o_t3/seed_NN.json). The outcome is strictly two-valued: 14/30 episodes at $F_1 = 1.0$ and 16/30 at $F_1 = 0.0$ (solve rate 0.467, Wilson 95% CI $[0.302, 0.639]$, consistent with the E1 estimate of 4/10). The system and task prompts are byte-identical across all 30 episodes, and every trajectory opens with the identical call `fetch_page(page=0)`, which the mock API’s 1-indexed paging (`page ≥ 1` validation) deterministically rejects with HTTP 422, delivered in the same `retry: true` envelope as the injected transient faults. Table A14 classifies every episode by its mechanically checkable transcript signature; the classifier is re-run (and asserted) at every regeneration of this appendix. No failing episode ever varies the page argument, emits a malformed submission call, or claims success: each re-issues the identical rejected call, announces it is giving up, and submits a schema-valid payload with an empty data section, a truthful `row_count` of 0, and a run log stating the data could not be fetched. Conversely, no solved episode mishandles the task content: all 14 submit exactly the 25 unique ground-truth records (duplicates removed, zero spurious rows).

Task	Paging	Page size	Max req.	Rows	QPS	Fault injection
T1_single_page	page	1000	5	800	5	none
T2_multi_page	page	500	50	2345	5	pagination
T3_duplicates	offset	10	50	25	5	duplicates: duplicate_rate=0.08, cross_page_duplicate_rate=0.03
T4_rate_limit_429	page	10	60	30	3	rate_limit: fail_on=[2]
T5_server_error_500	page	10	60	30	3	server_error: fail_on=[2]
T6_page_drift	page	12	60	36	5	page_drift
T7_totals_trap	offset	250	60	750	5	totals_trap
T8_mixed_faults	page	20	80	60	2	mixed: rate_limit_fail_on=[2, 5], duplicate_rate=0.1, cross_page_duplicate_rate=0.05
T9_adaptive_adversary	page	15	100	75	2	adaptive: initial_duplicate_rate=0.05, escalation_per_page=0.03, adaptive_429_after_page=3, adaptive_totals_after_page=4
T10_constrained_budget	page	10	50	80	5	duplicates: duplicate_rate=0.05, cross_page_duplicate_rate=0.02
T9_clean_ab	page	15	100	75	2	none

Table A11: Per-task parameters, generated from the released `server/tasks.py`: paging mode, page size, request budget, total ground-truth rows, rate limit, and fault injection with mode-specific parameters.

C.7 B2 and B2-Slack: Raw Per-Seed Coverage and Decisions

Escalating-arm coverage at zero slack is *identical* for Llama and GPT-5 on all 20 seeds (Claude differs on seed 18 only): the decision-equivalence disclosed in the main text, visible below rather than asserted. Decision columns count model choices across the 20 episodes of each arm.

C.8 Design Sensitivity: Monte-Carlo Power Study

Simulated power of the paired matched-mean test by effect regime and seeds per arm (400 replications per cell; paired bootstrap CI at $\alpha=0.05$ excluding zero as the decision rule). The pre-registered $n=20$ reaches near-ceiling power for large effects and the null-control regime confirms the test size.

C.9 Degradation Curves at Constant Intensity

Figure A5 plots coverage against *constant* adversary dose under the quota-enforced half-scaffold instrument (budget 8, no squeeze), 10 seeds per (model, level), from `results_degradation.json`. The per-seed spread below the mean opens as the dose rises (a mean–worst gap of zero for every model at dose 0, 0.12–0.21 at dose 1.0): the within-task counterpart of the suite-level mean–worst divergence in the main text.

C.10 Pre-Registration Digest and Artifact Provenance

The Claim B protocol was frozen before the confirmatory run (`PREREGISTRATION.md`): directional hypothesis (escalating worse), design (matched-mean arms, $n=20$ paired seeds, three models), decision rule (bootstrap CI excluding zero and sign-flip $p < 0.05$), and the explicit deferral of the enforced-quota variant as an *exploratory* ablation, the variant reported here as B2.

Disclosed deviation from the frozen document. The pre-registration specified the de-scaffolded slim agent with the terminal judge reward as the outcome; the confirmatory

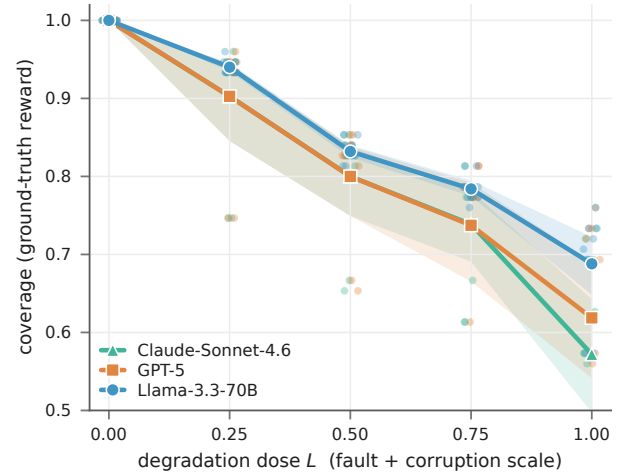


Figure A5: Degradation dose-response: per-seed coverage (dots), the mean (solid, with markers) and a seeded 95% bootstrap CI band, versus constant adversary dose under the quota-enforced half-scaffold instrument (no mid-episode squeeze); three models, 10 seeds per point. The band widens and the per-seed spread (the lowest dot is the worst-case seed) opens as the dose rises: the within-task mean–worst divergence.

run used the L1 half-scaffold agent with ground-truth fetch coverage. Because coverage deduplicates fetched records and filters totals rows before scoring, the pre-registered dedup/totals effect channel has no causal path to the outcome, and with retries free the only model-sensitive channel is an explicit SKIP, which fired in 0 of 120 episodes. Per the pre-registration’s own rule, the run is therefore reported as a null under an instrument that isolates a single decision, not as a powered test of the frozen hypothesis (Section 5 (main text)); the "agent" metadata

Model	Task	n	Mean	Std	Worst	CVaR	Rel@0.5
Claude-Fable-5	T3 dedup	10	1.000	0.000	1.000	1.000	1.000
Claude-Fable-5	T4 429-retry	10	1.000	0.000	1.000	1.000	1.000
Claude-Fable-5	T5 500-retry	10	1.000	0.000	1.000	1.000	1.000
Claude-Fable-5	T6 page-drift	10	1.000	0.000	1.000	1.000	1.000
Claude-Fable-5	T8 mixed	10	0.869	0.030	0.824	0.824	1.000
Claude-Fable-5	T7 totals	10	0.043	0.041	0.011	0.012	0.000
GPT-5	T3 dedup	10	1.000	0.000	1.000	1.000	1.000
GPT-5	T4 429-retry	10	1.000	0.000	1.000	1.000	1.000
GPT-5	T5 500-retry	10	1.000	0.000	1.000	1.000	1.000
GPT-5	T6 page-drift	10	0.806	0.017	0.800	0.800	1.000
GPT-5	T8 mixed	10	0.844	0.294	0.000	0.406	0.900
GPT-5	T10 budget	10	1.000	0.000	1.000	1.000	1.000
GPT-5	T7 totals	10	0.017	0.032	0.000	0.000	0.000
Claude-Sonnet-4.6	T3 dedup	10	1.000	0.000	1.000	1.000	1.000
Claude-Sonnet-4.6	T4 429-retry	10	1.000	0.000	1.000	1.000	1.000
Claude-Sonnet-4.6	T5 500-retry	10	1.000	0.000	1.000	1.000	1.000
Claude-Sonnet-4.6	T6 page-drift	10	0.800	0.000	0.800	0.800	1.000
Claude-Sonnet-4.6	T8 mixed	10	0.914	0.074	0.824	0.824	1.000
Claude-Sonnet-4.6	T7 totals	10	0.127	0.059	0.005	0.059	0.000
Claude-Haiku-4.5	T3 dedup	10	1.000	0.000	1.000	1.000	1.000
Claude-Haiku-4.5	T4 429-retry	10	1.000	0.000	1.000	1.000	1.000
Claude-Haiku-4.5	T5 500-retry	10	1.000	0.000	1.000	1.000	1.000
Claude-Haiku-4.5	T6 page-drift	10	0.799	0.004	0.787	0.793	1.000
Claude-Haiku-4.5	T8 mixed	10	0.824	0.000	0.824	0.824	1.000
Claude-Haiku-4.5	T7 totals	10	0.023	0.045	0.000	0.000	0.000
GPT-4o	T3 dedup	10	0.400	0.490	0.000	0.000	0.400
GPT-4o	T4 429-retry	10	1.000	0.000	1.000	1.000	1.000
GPT-4o	T5 500-retry	10	1.000	0.000	1.000	1.000	1.000
GPT-4o	T6 page-drift	10	0.800	0.000	0.800	0.800	1.000
GPT-4o	T8 mixed	10	0.824	0.000	0.824	0.824	1.000
GPT-4o	T10 budget	2	0.000	0.000	0.000	0.000	0.000
GPT-4o	T7 totals	10	0.026	0.017	0.000	0.000	0.000
GPT-4o-mini	T3 dedup	10	0.000	0.000	0.000	0.000	0.000
GPT-4o-mini	T4 429-retry	10	0.900	0.300	0.000	0.500	0.900
GPT-4o-mini	T5 500-retry	10	1.000	0.000	1.000	1.000	1.000
GPT-4o-mini	T6 page-drift	10	0.327	0.268	0.000	0.000	0.600
GPT-4o-mini	T8 mixed	10	0.824	0.000	0.824	0.824	1.000
GPT-4o-mini	T7 totals	10	0.000	0.000	0.000	0.000	0.000
Llama-3.3-70B	T3 dedup	10	0.000	0.000	0.000	0.000	0.000
Llama-3.3-70B	T4 429-retry	10	0.000	0.000	0.000	0.000	0.000
Llama-3.3-70B	T5 500-retry	10	0.000	0.000	0.000	0.000	0.000
Llama-3.3-70B	T6 page-drift	10	0.000	0.000	0.000	0.000	0.000
Llama-3.3-70B	T8 mixed	10	0.000	0.000	0.000	0.000	0.000
Llama-3.3-70B	T10 budget	10	0.000	0.000	0.000	0.000	0.000
Qwen2.5-7B	T3 dedup	10	0.000	0.000	0.000	0.000	0.000
Qwen2.5-7B	T4 429-retry	10	0.000	0.000	0.000	0.000	0.000
Qwen2.5-7B	T5 500-retry	10	0.000	0.000	0.000	0.000	0.000
Qwen2.5-7B	T6 page-drift	10	0.000	0.000	0.000	0.000	0.000
Qwen2.5-7B	T8 mixed	10	0.000	0.000	0.000	0.000	0.000

Table A12: E1 complete per-(model, task) statistics from `results_reliability_full.json`, including partial runs and uneven seed counts.

field of `results_t9_ab_full.json` has been corrected from `slim` to `half_scaffold_coverage`. A pre-registration-faithful rerun (slim agent, judge-reward outcome, identical adversary; `harness_b1_slim.py` $\rightarrow$ `results_b1_slim.json`) has been completed for all three pre-registered models and is reported in Section 5 (main text): under the frozen decision rule GPT-5 confirms on the

terminal judge reward while its ground-truth and coverage deltas are ≈ 0 . Artifact map:

- E1: `nonstationary/harness_reliability_full.py` $\rightarrow$ `results_reliability_full.json`

- B1 (pre-registered protocol, deviating instrument as disclosed above): `harness_t9_ab_full.py` → `results_t9_ab_full.json`
- B1 pre-registration-faithful re-run: `harness_b1_slim.py` → `results_b1_slim.json`
- External scorer probe: `external/taubench_probe.py` → `taubench_probe_results.json`
- External scaffold-variance slice: `external/taubench_slice/` (run `run_grid.sh`, `run_fewshot.sh` → per-cell result JSONs; Table A3)
- B2 / B2-slack (exploratory): `harness_b2.py` → `results_b2_squeeze.json`, `results_b2_slack.json`, `results_b2_slack_gpt5.json`
- B2 rule-agent calibration (no-LLM bound): `harness_b2_rule.py` → `results_b2_rule.json`
- GPT-4o T3 failure forensics ($n=30$, full transcripts; Section C.6): `harness_gpt4o_forensics.py` → `results_gpt4o_t3_n30.json`, `forensics_gpt4o_t3/seed_NN.json`
- Degradation curves: `harness_degradation.py` → `results_degradation.json`
- T7/T8 completion runs: `harness_reliability_full.py` with AB_TASKS/AB_MODELS knobs plus the per-seed resumer `resume_t7_seeds.py` → `results_t7t8_anthropic.json`, `results_t7t8_openai.json`, merged by `merge_t7t8.py` (refuses incomplete coverage) into `results_reliability_full.json`
- Judge probe: `paper/aaai27/probe_judge.py` → `probe_judge_results.json` (Table A10), against the released server/judge.py in the code release
- Figures: `Figures/figure1_regen.py` regenerates the pipeline/spectrum figure from `llm_results_{llama,gpt5,claude}.json`, `results_t9_ab_full.json`, and `results_reliability_full.json`; `Figures/figure2.py` regenerates the escalation-forest figure from the B1/B2 files above; `Figures/figure4_degradation.py` regenerates Figure A5 from `results_degradation.json`; `Figures/figure6_spectrum.py` regenerates the eight-model E1 spectrum (Figure A9) from `results_reliability_full.json`; `Figures/figure7_judgeprobe.py` regenerates the judge-probe bars (Figure A8) from `probe_judge_results.json`; `Figures/figure8_interaction.py` regenerates the scaffold-model interaction chart from the per-episode JSONs in `external/taubench_slice/`; `Figures/figure9_confound.py` draws the two-path confound schematic (no data inputs).

D Complete Figure Record

This appendix is the complete figure record of the project. It has two parts. First, the measurement figures: the scaffolding-spectrum figure, which places the three models present at every scaffolding level side by side (plus Claude-Fable-5 at the one level it was run) and discloses the provenance of every plotted number, and the eight-model E1 reliability spectrum, the visual companion of main-text Figure 2. Second, the legacy artifacts of the original benchmark release: the judge-scored, production-scaffolded (L0) charts and the GRPO (Shao et al. 2024) training figures produced against the original reward. Nothing here is re-scored or re-drawn; the legacy figures are preserved exactly as released, because under the paper’s diagnosis they are primary evidence: their flat leaderboards are the double measurement confound in action.

D.1 The 2×2 Joint Intervention

Figure A6 draws main-text Table 1 as four panels, one for each cell of the harness-by-scorer swap.

D.2 The Scaffolding Spectrum

Figure A7 plots the values in Table A19. Provenance of each column:

- **L0 (production scaffold).** Mean of the 10 per-task score values, /100, from `llm_results_{llama,gpt5,claude}.json`. Each mean matches the file’s own `average_score` field (89.3, 93.2, 97.5).
- **L1 (half-scaffold).** `summary.cm_mean` (the constant arm) from `results_t9_ab_full.json`. The escalating-arm mean is also 0.832 for all three models, so the mean over both arms is identical to the value shown.
- **L2 (de-scaffolded).** Mean over the 50 per-run rewards of T3_duplicates, T4_rate_limit_429, T5_server_error_500, T6_page_drift, and T8_mixed_faults from `results_reliability_full.json`, the same five fully-run execution tasks scoped by main-text Figure 2 (identical to its Mean column). Claude-Fable-5 has no L0/L1 entries: it was added to the suite at E1 (L2) time and never run under the production or half scaffold.

Disclosed deviation from the figure brief. The brief anticipated L0 spanning roughly 0.93–0.975, but Llama-3.3-70B’s measured L0 is 0.893 (its released `average_score` is 89.3). The measured value is plotted unchanged; no number was adjusted to match the brief. The figure was visually verified after generation (no clipping; axis, legend, and per-level metric sublabels legible), and both .pdf and .png renders are produced by the same script.

D.3 The Judge Probe, Visualized

Figure A8 plots Table A10 as bars, so the floor and the tie can be read off at a glance.

D.4 The Eight-Model E1 Reliability Spectrum

Figure A9 is the visual companion of main-text Figure 2: one bar and one worst-case marker per model.

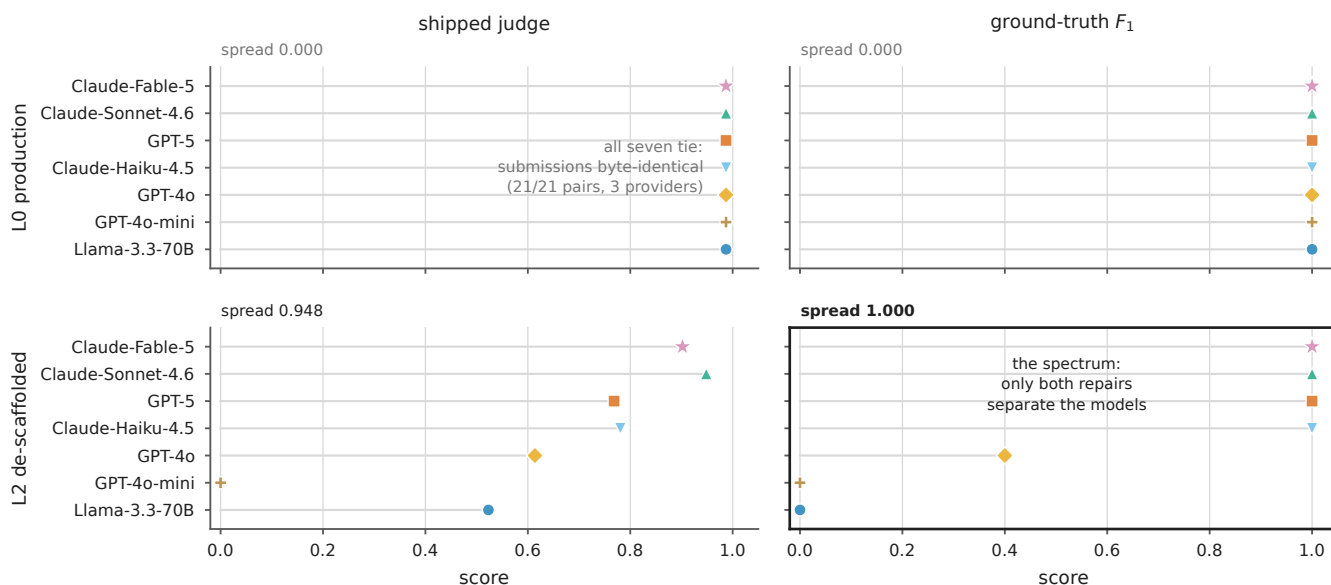


Figure A6: The joint intervention of main-text Table 1 as a picture: rows are the harness swap, columns are the scorer swap, and each panel plots the same seven models on the same axis (values machine-read from the released `joint_matrix.json`). Top row: the shipped condition; seven models from three providers collapse to one vertical line under either scorer because the scaffold’s submissions are byte-identical. Bottom left: the judge separates only failure-to-submit and misorders what it grades (Sonnet above the ground-truth-perfect GPT-5). Bottom right: model-owned execution against ground truth recovers the spectrum.

D.5 Legacy Artifacts: the Original Benchmark Release

figures), with the original release renders preserved in `Figures/legacy/`.

The figures below are the artifacts of the original release of the benchmark (Zhang et al. 2026) and its accompanying report. All benchmark numbers in them were produced by the *original six-dimension judge under the production scaffold*, the L0 configuration whose confound Section 3 (main text) diagnoses. We reproduce them without alteration, for three reasons. First, completeness: this is the full figure record, not a curated subset. Second, the flat leaderboards visible in Figures A10 and A12 are not a historical curiosity but the confound in action: a no-LLM rule-based script sits inside the model pack, and models of very different capability tie to within a few judge points on T1–T8. Third, the GRPO figures (Figures A13–A15) document a training study run against the original L0 judge reward that is not discussed in the main text; we preserve it because the observed saturation-at-initialization and reward-variance collapse are themselves evidence that the L0 reward carries little model-discriminating signal: a policy can sit at the reward ceiling before a single gradient step.

Where a figure was released in both vector and raster form (Figures A11, A12, A13, A16) we render the PDF; the PNG render of each ships alongside it in `Figures/legacy/`. The benchmark-results chart (Figure A10) exists only as PNG; the two GRPO figures (Figures A14, A15) are re-generated here in print form directly from the raw per-iteration training records (generating scripts alongside the

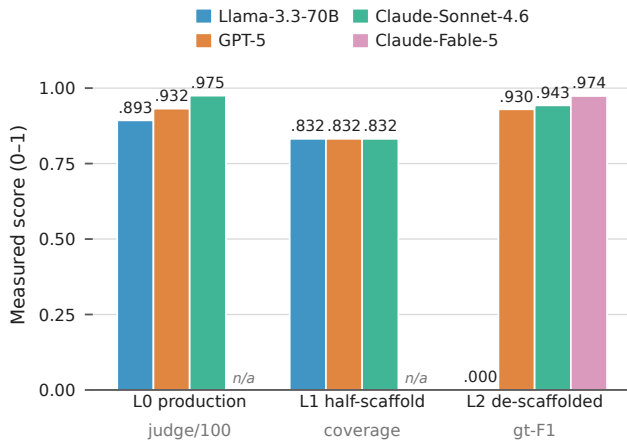


Figure A7: The scaffolding spectrum: measured performance of the three models present at every scaffolding level, plus Claude-Fable-5 at the one level it was run (L2; its L0/L1 slots read *n/a*, absence rather than zero). The metric changes with the level, and that is the point: L0 (production scaffold) is the original six-dimension judge score, /100 normalized to $[0, 1]$; L1 (half-scaffold) is coverage; L2 (de-scaffolded) is ground-truth F_1 averaged over the 50 runs of the five execution tasks. L0 and L1 are flat or near-flat across models; only L2 separates them. Bars compare models *within* a level; the three metrics are not commensurable *across* levels. Every value is read programmatically from the released result JSONs by `Figures/figure3_scaffolding.py`; nothing is hand-typed.

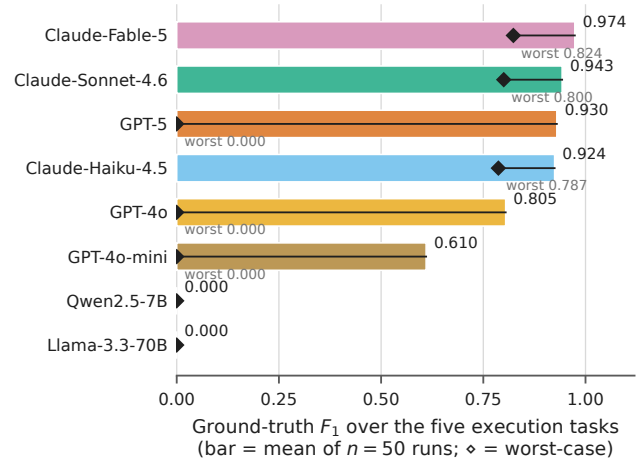


Figure A9: The eight-model E1 reliability spectrum, the visual companion of main-text Figure 2: mean ground-truth F_1 (bar) and worst-case over all 50 runs (diamond) per model at L2 on the five execution tasks, sorted by mean. The bar-to-diamond distance is the reliability gap. Claude-Fable-5 tops both mean and worst case without saturating; GPT-5's gap (0.93, a mixed-fault hard zero) is now the suite's largest: the worst-case reorder of the main text, reaching the frontier. Every value is computed from `results_reliability_full.json` by `Figures/figure6_spectrum.py` (mean over the 50 per-run rewards; worst = minimum); nothing is hand-typed, and the values match main-text Figure 2 exactly.

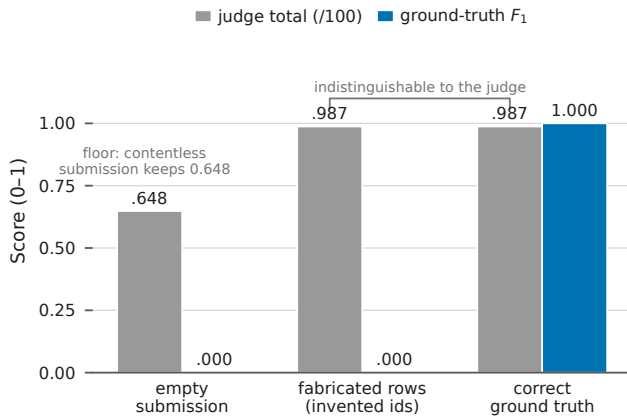


Figure A8: The judge probe (Table A10) as bars: three canned submissions through the released, unmodified `judge.py`, each scored by the judge total (/100, gray) and by ground-truth F_1 (blue). The two scorer artifacts are visible at a glance: a contentless submission is floored at 0.648, and the fabricated record set is indistinguishable from the correct one to the judge (0.987 each) while ground-truth F_1 separates them at 0.000 versus 1.000. Values read from `probe_judge_results.json` by `Figures/figure7_judgeprobe.py`; nothing is hand-typed.

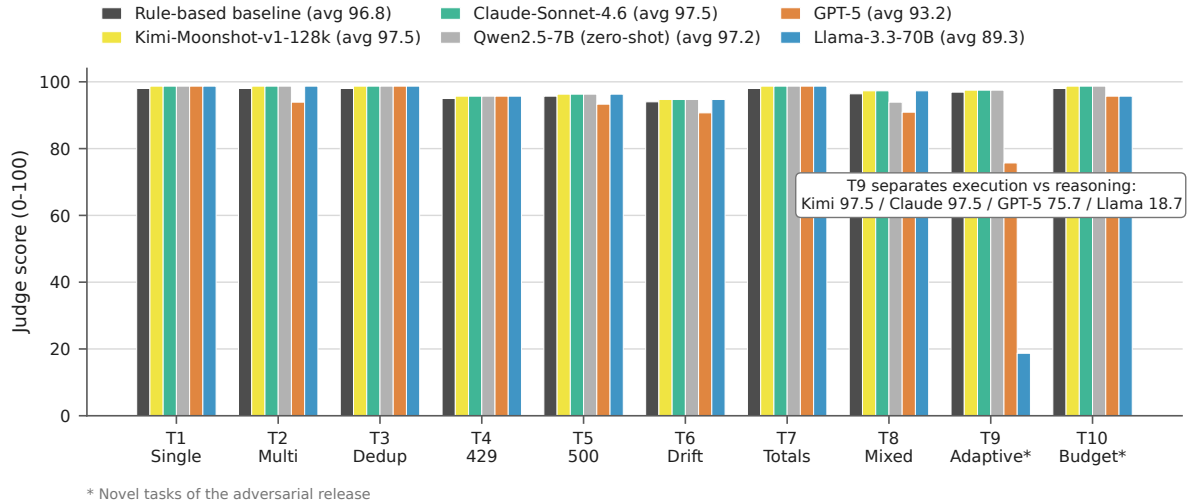


Figure A10: Cross-model results on the original 10-task suite (T1–T10), redrawn in this paper’s figure style from the original release’s per-run score files (data unchanged; the original chart is preserved in the repository). Scorer: original six-dimension judge, 0–100; scaffold: L0 production. Bars are per-task judge scores for six systems, including a no-LLM rule-based baseline. Bars are near-identical across systems on every task except the T9 adaptive adversary (and the rule-based script sits inside the model pack): the flat-leaderboard signature of scoring the scaffold rather than the model.

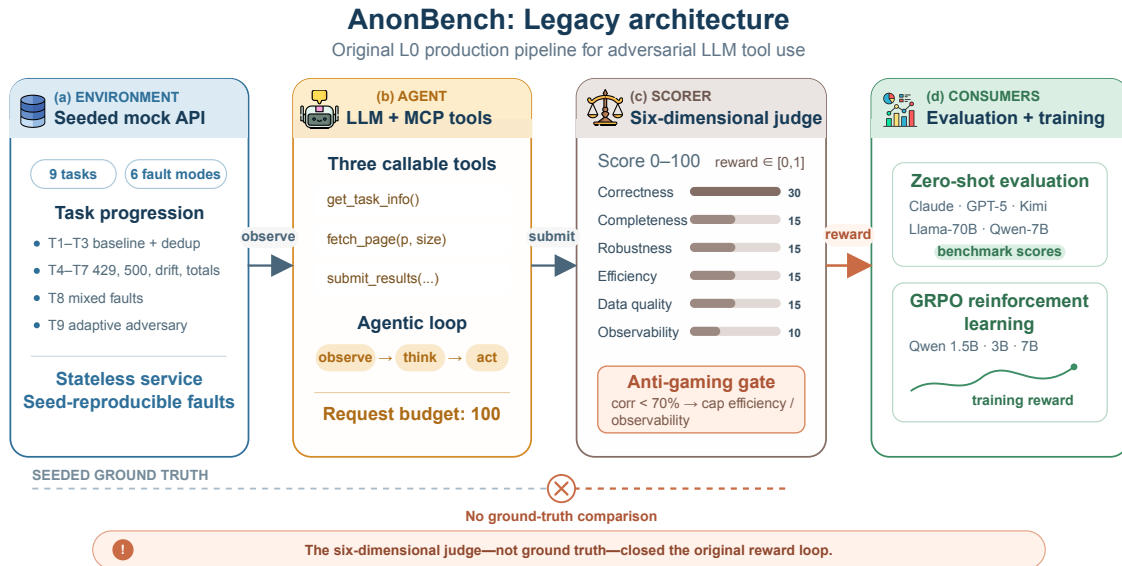


Figure A11: Legacy architecture schematic of the original release (its footer credit line was cropped in the review version; the uncropped original ships in the artifact bundle): (a) the mock trade-statistics API serving nine seeded tasks with synthetic fault modes, (b) the LLM agent loop over three MCP tools under a request budget, (c) the six-dimension judge with anti-gaming gates producing the 0–100 score used as reward, and (d) the zero-shot evaluation and GRPO training consumers. Panel (c) is the L0 scorer diagnosed in the main text; the schematic documents that the judge, not ground truth, closed the original reward loop.

Model	Task	Per-seed ground-truth F_1
Claude-Fable-5	T3 dedup	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
Claude-Fable-5	T4 429-retry	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
Claude-Fable-5	T5 500-retry	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
Claude-Fable-5	T6 page-drift	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
Claude-Fable-5	T8 mixed	[0.823529, 0.888889, 0.823529, 0.888889, 0.888889, 0.888889, 0.823529, 0.888889, 0.888889, 0.888889]
Claude-Fable-5	T7 totals	[0.0288714, 0.106061, 0.010596, 0.015852, 0.106061, 0.0132275, 0.106061, 0.015852, 0.015852, 0.015852]
GPT-5	T3 dedup	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
GPT-5	T4 429-retry	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
GPT-5	T5 500-retry	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
GPT-5	T6 page-drift	[0.8, 0.8, 0.8, 0.8, 0.8, 0.8, 0.8, 0.8, 0.8, 0.857143]
GPT-5	T8 mixed	[1, 1, 1, 1, 0, 0.811881, 1, 1, 0.811881, 0.811881]
GPT-5	T10 budget	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
GPT-5	T7 totals	[0, 0, 0.0262812, 0, 0, 0, 0.103535, 0, 0, 0.0366492]
Claude-Sonnet-4.6	T3 dedup	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
Claude-Sonnet-4.6	T4 429-retry	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
Claude-Sonnet-4.6	T5 500-retry	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
Claude-Sonnet-4.6	T6 page-drift	[0.8, 0.8, 0.8, 0.8, 0.8, 0.8, 0.8, 0.8, 0.8, 0.8]
Claude-Sonnet-4.6	T8 mixed	[0.823529, 0.974359, 0.974359, 0.974359, 0.974359, 0.823529, 0.823529, 0.974359, 0.823529, 0.974359]
Claude-Sonnet-4.6	T7 totals	[0.113208, 0.00531915, 0.192771, 0.113208, 0.148148, 0.113208, 0.113208, 0.113208, 0.113208, 0.24356]
Claude-Haiku-4.5	T3 dedup	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
Claude-Haiku-4.5	T4 429-retry	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
Claude-Haiku-4.5	T5 500-retry	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
Claude-Haiku-4.5	T6 page-drift	[0.8, 0.8, 0.8, 0.8, 0.8, 0.8, 0.786885, 0.8, 0.8, 0.8]
Claude-Haiku-4.5	T8 mixed	[0.823529, 0.823529, 0.823529, 0.823529, 0.823529, 0.823529, 0.823529, 0.823529, 0.823529, 0.823529]
Claude-Haiku-4.5	T7 totals	[0, 0.113208, 0, 0, 0, 0.113208, 0, 0, 0, 0]
GPT-4o	T3 dedup	[0, 0, 1, 0, 0, 0, 0, 1, 1, 1]
GPT-4o	T4 429-retry	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
GPT-4o	T5 500-retry	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
GPT-4o	T6 page-drift	[0.8, 0.8, 0.8, 0.8, 0.8, 0.8, 0.8, 0.8, 0.8, 0.8]
GPT-4o	T8 mixed	[0.823529, 0.823529, 0.823529, 0.823529, 0.823529, 0.823529, 0.823529, 0.823529, 0.823529, 0.823529]
GPT-4o	T10 budget	[0, 0]
GPT-4o	T7 totals	[0, 0.0366013, 0, 0.0366013, 0, 0.0366013, 0.0366013, 0.0366013, 0.0366013, 0.0366013]
GPT-4o-mini	T3 dedup	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
GPT-4o-mini	T4 429-retry	[0, 1, 1, 1, 1, 1, 1, 1, 1, 1]
GPT-4o-mini	T5 500-retry	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
GPT-4o-mini	T6 page-drift	[0, 0, 0.530612, 0.530612, 0.530612, 0.615385, 0, 0.530612, 0, 0.530612]
GPT-4o-mini	T8 mixed	[0.823529, 0.823529, 0.823529, 0.823529, 0.823529, 0.823529, 0.823529, 0.823529, 0.823529, 0.823529]
GPT-4o-mini	T7 totals	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
Llama-3.3-70B	T3 dedup	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
Llama-3.3-70B	T4 429-retry	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
Llama-3.3-70B	T5 500-retry	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
Llama-3.3-70B	T6 page-drift	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
Llama-3.3-70B	T8 mixed	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
Llama-3.3-70B	T10 budget	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
Qwen2.5-7B	T3 dedup	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
Qwen2.5-7B	T4 429-retry	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
Qwen2.5-7B	T5 500-retry	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
Qwen2.5-7B	T6 page-drift	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
Qwen2.5-7B	T8 mixed	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]

Table A13: E1 raw per-seed vectors (seed order 0,1,...).

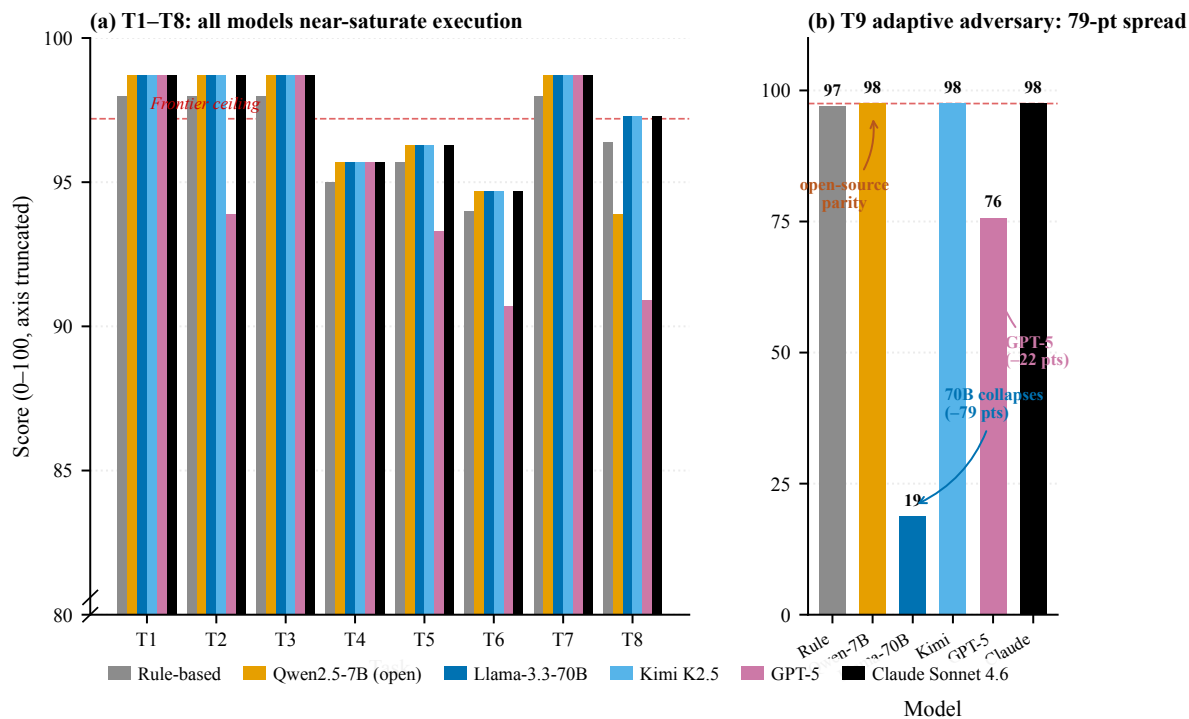


Figure A12: Legacy Figure 1 of the original release (vector version): two-panel benchmark chart, scored by the original six-dimension judge under the L0 production scaffold. Panel (a): T1-T8 judge scores for six systems (rule-based baseline included) on a truncated 80-100 axis, all crowding a “frontier ceiling” line, near-saturated execution. Panel (b): the T9 adaptive-adversary task, the one task in the original suite that spreads the systems, with an annotated 79-point spread. The original release read panel (b) as the interesting result; this paper’s diagnosis is that panel (a) is the finding.

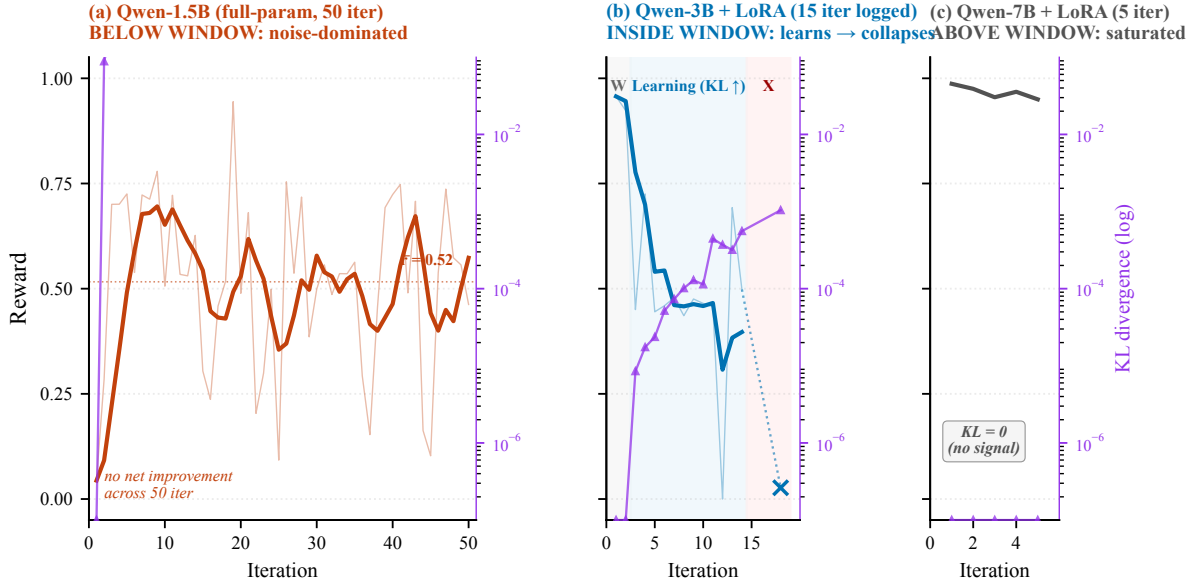


Figure A13: Legacy Figure 2 of the original release (vector version): GRPO training dynamics against the original L0 judge reward, for three model scales. Panel (a): Qwen-1.5B full-parameter, 50 iterations; reward oscillates with no net trend (below the trainable window; noise-dominated). Panel (b): Qwen-3B+LoRA (Hu et al. 2022); reward and KL divergence move during a learning phase, then the run collapses. Panel (c): Qwen-7B+LoRA, 5 iterations; reward starts at ceiling with $KL = 0$ (no gradient signal; saturated). Not discussed in the main text; preserved because saturation-at-initialization under the L0 reward is direct evidence that the reward carries little model-discriminating signal.

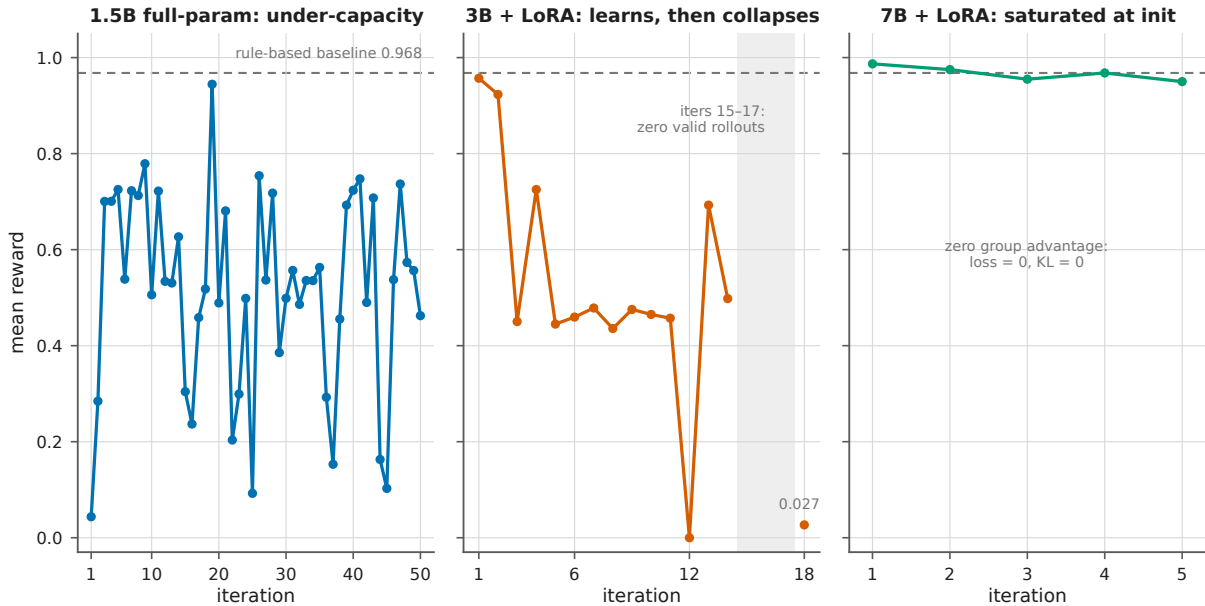


Figure A14: GRPO operating envelope, regenerated from the raw per-iteration records: mean reward per iteration under the original L0 judge reward for the same three runs as Figure A13, against the rule-based baseline (0.968, dashed). The three empirically mapped failure modes: under-capacity oscillation (1.5B full-param), learns-then-collapses (3B+LoRA; iterations 15–17 produced zero valid rollouts, shaded, before the final recorded point at 0.027), and saturated-at-init (7B+LoRA, near-zero reward variance, hence zero GRPO advantage). The narrowness of the usable band is a property of the reward, not of GRPO.

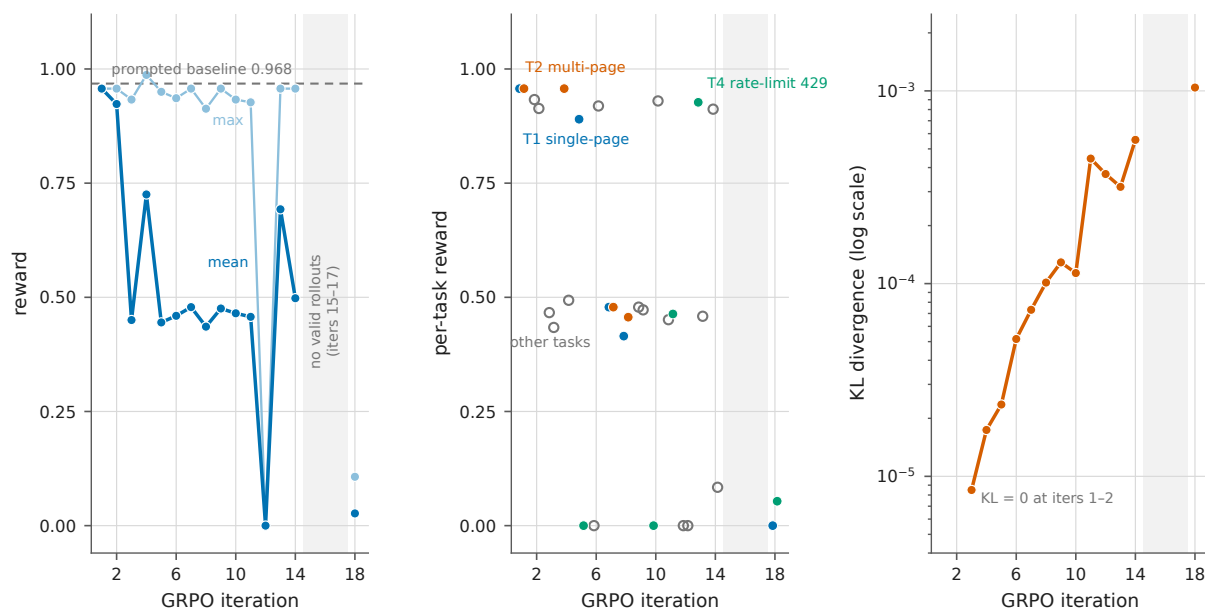


Figure A15: GRPO training metrics for the 3B+LoRA run, regenerated from the raw per-iteration records: mean and max reward per iteration (left), per-task reward scatter (middle; the three most frequent tasks direct-labeled, remaining tasks in gray), and KL divergence on a log scale (right; loss is omitted; it oscillates around zero and carries no printable trend on a single axis). Reward is the original six-dimension judge score (L0), normalized. KL grows monotonically through the collapse (the adapter keeps moving *through* the degenerate region), and the per-task panel shows the reward arriving as sparse task spikes rather than a graded curriculum: context for Figures A13 and A14.

Transcript signature	<i>n</i>	Ex. seed
Solved: probes page 1 after 3–4 rejections of page 0, fetches pages 1–3, submits 25 unique rows	14	1
Failed: 4 identical <code>page=0</code> calls, then schema-valid empty submission declaring fetch failure	12	0
Failed: extended perseverance (5–9 identical calls), same terminal capitulation	4	17
Failed: malformed submission, or success claimed without data	0	n/a

Table A14: Mechanism taxonomy over all 30 forensic episodes, computed by re-parsing every transcript in `forensics_gpt4o_t3/`. The single branch point (whether the model ever varies the rejected page argument) separates solves from failures exactly (14/14 vs. 0/16).

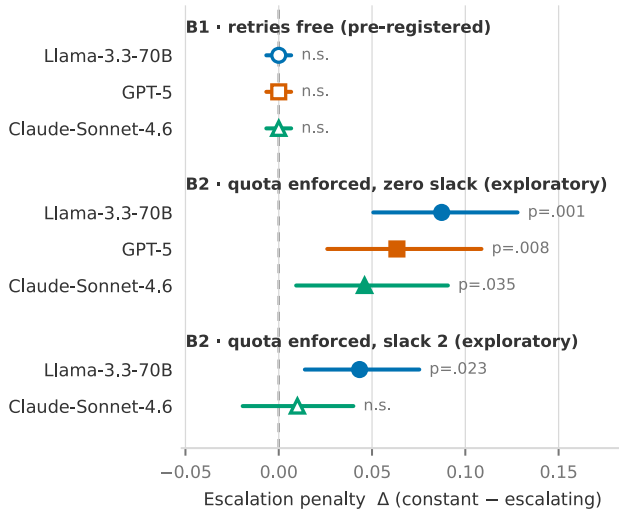


Figure A16: Legacy copy of the paper’s escalation-penalty forest plot (vector version): paired difference Δ (constant – escalating) in coverage per model, with bootstrap intervals, for the pre-registered B1 null (retries free; all n.s.), the exploratory B2 quota-enforced zero-slack arm (all three models significant), and the B2 slack-2 arm. Metric: coverage under the L1 half-scaffold harness, not the judge. Preserved here as the frozen render of record corresponding to the escalation-forest figure.

Model	Arm	Per-seed coverage	Retries	Skips	Quota deaths
GPT-5	escalating	[0.53, 0.68, 0.84, 0.81, 0.68, 0.68, 0.83, 0.68, 0.85, 0.83, 0.68, 0.71, 0.68, 0.71, 0.83, 0.68, 0.68, 0.83, 0.68, 0.84]	8	9	8
GPT-5	constant@mean	[0.85, 0.83, 0.84, 0.83, 0.83, 0.65, 0.84, 0.67, 0.85, 0.81, 0.84, 0.83, 0.64, 0.85, 0.81, 0.85, 0.81, 0.83, 0.68, 0.84]	13	4	0
Claude-Sonnet-4.6	escalating	[0.53, 0.68, 0.84, 0.81, 0.68, 0.68, 0.83, 0.68, 0.85, 0.83, 0.68, 0.71, 0.68, 0.71, 0.83, 0.68, 0.68, 0.83, 0.52, 0.84]	6	11	7
Claude-Sonnet-4.6	constant@mean	[0.85, 0.83, 0.84, 0.83, 0.83, 0.65, 0.84, 0.67, 0.85, 0.81, 0.84, 0.83, 0.64, 0.68, 0.81, 0.85, 0.65, 0.83, 0.51, 0.84]	10	7	0
Llama-3.3-70B	escalating	[0.53, 0.68, 0.84, 0.81, 0.68, 0.68, 0.83, 0.68, 0.85, 0.83, 0.68, 0.71, 0.68, 0.71, 0.83, 0.68, 0.68, 0.83, 0.68, 0.84]	11	6	11
Llama-3.3-70B	constant@mean	[0.85, 0.83, 0.84, 0.83, 0.83, 0.81, 0.84, 0.83, 0.85, 0.81, 0.84, 0.83, 0.80, 0.85, 0.81, 0.85, 0.81, 0.83, 0.68, 0.84]	16	1	0

Table A15: B2 raw record, zero slack (budget 8 $\rightarrow$ 5), from `results_b2_squeeze.json`.

Model	Arm	Per-seed coverage	Retries	Skips	Quota deaths
Claude-Sonnet-4.6	escalating	[0.71, 0.68, 0.84, 0.81, 0.68, 0.83, 0.83, 0.68, 0.85, 0.83, 0.83, 0.85, 0.68, 0.85, 0.83, 0.83, 0.68, 0.83, 0.68, 0.84]	12	7	0
Claude-Sonnet-4.6	constant@mean	[0.85, 0.83, 0.84, 0.83, 0.83, 0.65, 0.84, 0.67, 0.85, 0.81, 0.84, 0.83, 0.64, 0.85, 0.81, 0.85, 0.65, 0.83, 0.68, 0.84]	12	5	0
Llama-3.3-70B	escalating	[0.71, 0.68, 0.84, 0.81, 0.68, 0.83, 0.83, 0.68, 0.85, 0.83, 0.83, 0.85, 0.83, 0.85, 0.83, 0.83, 0.68, 0.83, 0.68, 0.84]	16	5	2
Llama-3.3-70B	constant@mean	[0.85, 0.83, 0.84, 0.83, 0.83, 0.81, 0.84, 0.83, 0.85, 0.81, 0.84, 0.83, 0.80, 0.85, 0.81, 0.85, 0.81, 0.83, 0.85, 0.84]	17	0	0

Table A16: B2 raw record, slack 2 (budget 10 $\rightarrow$ 7), from `results_b2_slack.json`.

Model	Arm	Per-seed coverage	Retries	Skips	Quota deaths
GPT-5	escalating	[0.71, 0.83, 0.84, 0.81, 0.68, 0.83, 0.83, 0.68, 0.85, 0.83, 0.83, 0.85, 0.68, 0.85, 0.83, 0.83, 0.68, 0.83, 0.68, 0.84]	14	6	1
GPT-5	constant@mean	[0.85, 0.83, 0.84, 0.83, 0.83, 0.65, 0.84, 0.67, 0.85, 0.81, 0.84, 0.83, 0.64, 0.85, 0.81, 0.85, 0.81, 0.83, 0.85, 0.84]	14	3	0

Table A17: B2 raw record, slack 2 (budget 10 $\rightarrow$ 7), GPT-5 run, from `results_b2_slack_gpt5.json`.

Regime	True Δ	$n=3$	$n=5$	$n=8$	$n=12$	$n=20$	$n=30$	$n=50$
null control	-0.000	0.08	0.07	0.07	0.05	0.09	0.06	0.04
small effect	+0.032	0.11	0.12	0.10	0.14	0.18	0.26	0.31
medium effect	+0.058	0.15	0.23	0.21	0.33	0.39	0.57	0.70
large effect	+0.209	0.48	0.74	0.84	0.92	1.00	1.00	1.00

Table A18: Power by effect regime and n per arm.

Model	L0 (judge)	L1 (cov.)	L2 (F_1)
Llama-3.3-70B	0.8932	0.8320	0.0000
GPT-5	0.9320	0.8320	0.9299
Claude-Sonnet-4.6	0.9750	0.8320	0.9428
Claude-Fable-5	n/a	n/a	0.9739

Table A19: Exact values plotted in Figure A7 (model $\times$ scaffolding level): L0 judge score /100, L1 coverage, L2 ground-truth F_1 over the five execution tasks (mean of the 50 per-run rewards). Claude-Fable-5 was run at L2 only.